MANUAL

ON

INTEGRATED FARMING SYSTEMS

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PREFACE

The glory of Green Revolution proved to be a turning point in improving the food situation from “begging bowl to self-sufficiency”. The future of Indian agriculture depends on the development of appropriate farming system as applicable to resource poor farm families and as suited to different agro-ecological zones. A judicious mix of cropping systems with associated enterprises like dairy, poultry, piggery, fishery, sericulture, bee keeping, mushroom production etc suited to the given agro- climatic conditions and socio-economic status of the farmers would bring prosperity to the farmer. Integrated farming system is not only a reliable way of obtaining fairly high productivity with substantial fertilizer economy, but also a concept of ecological soundness leading to sustainable agriculture.

The topics covered in the manual are most relevant in view of growing interests in integrated farming systems. The manual brings out in detail the agronomy course on integrated farming systems recommended for undergraduate students by 4th Dean’s Committee and accepted by the ICAR. The chapters included in this manual are a mixture of theoretical and applied investigation. The focus is on farming system definition, scope and characteristics; historical development of farming systems in India; concepts and components of farming system; cropping system, complementary and competitive interaction; indices for evaluation for cropping system; agronomic requirements in management of cropping system; role of farming systems in sustainable agriculture; models of integrated farming systems for irrigated ecosystems and rainfed ecosystems; importance and role of IFS’s in organic farming, low input sustainable agriculture and low cost agricultural technologies. The manual also covers the practical part of the course. All the exercises as per the course catalogue have been described in detail.

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THEORY

Chapter 1. Farming System, Definition, Classification, characteristics and Scope

Nature is dynamic. It causes inheritable changes in all living organisms. For thousands of years, farmer around the world have been selecting and conserving varieties of different crop plants that they cultivated. This process has generated a rich wealth of varieties in each crop plant, seen to be most abundant in countries near the equator and India is no exception. In the world there are Mega centre's of biodiversity of which were present in India. This is because in India, the different soils and agro-climatic situations are present. So, Indian farmers grow a large number of crops. Generation of Indian farmers, with their continued selection and conservation, has created a rich wealth of varieties in many crops. Therefore India is the original home of many crops such as rice, little and kodo millets, red gram, moth bean, jute, pepper, cardamom, many vegetables and fruit species. These plants were identified from the wild, selected and cultivated by Indian farmers over hundreds of years. The present wealth of varieties in India includes both crops that originated in the country and those that were introduced from the other countries during the distant and recent past like soybean, sunflower, oil pail and kiwi fruit.

The Indian economy is predominantly rural and agriculture oriented. In agriculture, 85% of the holdings are less than two hectares and the declining trend in the average size of the farmer holdings, poses a serious problem. Majority of them are dry lands, which depend on erratic monsoon rains. The rest of the area is cultivated with supplemental irrigation. The farmers concentrate mainly on crop production, which is invariably subjected to a high degree of uncertainly income and employment.

In India the cultivable land is 143.8 million hectares and there is very little possibility of extending it further. Therefore, to meet the requirement of food grains for increasing population, the only option open is through time and effective space utilization in agriculture. The time concept relates to increasing the intensity of cropping under assured irrigated conditions, whereas space utilization pertains to building up of vertical dimension through multi-tier cropping and farming system

approach. Thus, by making use of these time and space concept either in irrigated or in rained areas, the productivity per unit area per unit time can be substantially enhanced. Therefore the only way to increase an agricultural production in the small/marginal units of farming is to increase the productivity per unit time and area. This may be achieved of quicker maturing varieties with equal yields or by improving techniques of culture, fertilizer use, weed and pest control.

The green revolution in mid sixties stirred by research based new technological developments involving new materials, models and ways of organizing farm inputs (water, fertilizer, chemical etc) and government's policies transformed the agriculture dramatically. The outcome achiewed a many fold increase in production and productivity viz. food grain production of 260 million tons (which was only 74.23 million tons in 1966-67) and food grain productivity of 1800 kg/ha (which was only 644 kg/ha in 1966-67). The country thus became self sufficient in food production despite tremendous pressure to sustain 16% of world human population and 10% of cattle population with just 2.4% of total land.

However modern crop production technology has considerably raised output but has residues in farm produce, gene erosion of fossil fuel based inputs such as fertilizer, pesticide, herbicides and labour spacing but energy intensive farm machinery.

Land water and plant (bio diversity) are the natural resources and conservation of these natural resources is vital important. As far as resources degradation is concentrated, India is facing top soil depletion, falling ground water table and contamination of water, ecological imbalance due to deforestation and healthy hazards linked to increase use of toxic agro- chemical because of increase in food consumption. The adverse soil loss is estimated to be over 16 tons/ha/year, which is layer of top soil. Annual top soil loss is 6000 million tons, which carries about 8.4 million tones of nutrients and it is more than what we are using today. Similarly Tata Energy Research Institute (TERI) reported that more than 10% of GDP is being lost annually on account of environment and around 11-26% of country's apicultural output is lost on account of soil degradation.

The rate of human induced land degradation is very high. Out of total geographical area of 329 million ha, 187 million ha, representing almost 57% of total geographical area of the country are reported to have degraded of which 162 million ha is by way of wind and water related degradation.

What is Farming System?

"Farming system" therefore, designates a set of agricultural activities organized while preserving land productivity, environmental quality and maintaining desirable level of biological diversity and ecological stability. The emphasis is more on a system rather than on gross output.

In other words "farming system" is a resource management strategy to achieve economic and sustain agricultural production to meet diverse requirement of the farm household while preserving the resource base and maintaining high environmental quality. The farming system in its real sense will help the following ways to lift the economy of Indian agriculture and standard of living of the farmers. Farming system specially refers to a group combination of enterprises in which the products and or the byproducts of one enterprise serve as the inputs for production of other enterprise.

Farming system takes into account the combined needs of the family, the economic factors like relative profitability of the technically feasible enterprises, availability off farm resources, infrastructure and institutions such as irrigation, marketing facilities including storage and transportation and credit besides the agro biological consideration namely interdependence, if any among various technically feasible enterprises and the performance of individual farmers.

Farming is defined as the way in which the farm resources are allocated to the needs and priorities of the farmers in his local circumstances which include-

1. Agro climatic condition such as the quantity, distribution and reliability of rainfall. Soil type and topography temperature etc and

2. Economic and institutional circumstances like market opportunities, prices, institutional and infrastructure facilities and technology.

Definition:

Farming system is a decision making unit comprising the farm household, cropping and livestock system that transform land, capital and labour into useful products that can be consumed or sold.

Farming system is a resource management strategy to achieve economic and sustained production to meet diverse requirement of farm household while presenting resources base and maintaining a high level environmental quality (Lal and Millar 1990). They interact adequately with environment without dislocating the ecological and socio- economic balance on the one hand and attempt to meet the national goal on the other.

Farming system consist of several enterprises like cropping system, dairying, piggery, poultry, fishery, beekeeping etc. these enterprises are interrelated. The end product and wastes of one enterprise are used as inputs in others. The waste of dairying like dung, urine, refuse etc. is used for preparation of FYM, which is an input in cropping systems. The straw obtained from the crops is used as fodder for cattle, which are used for different field operations for growing crops. Thus different enterprises of farming system are highly interrelated.

Farming system is complexes inter related matrix of soil, plants, animals, implements, power, labour, capital and other inputs controlled in parts by farming families and influenced to varying degree by political, economic, institutional and social forces that operate at many levels. Thus farming system is the result of a complex interaction among a number of interdependent components. To achieve it, the individual farmer allocates four factors of production i.e. Land, labour, capital and management, which has access to processes like management which has crop, livestock and off farm enterprises in a manner, which within the knowledge he possess will maximize the attainment of goal he is striving for. Income through

arable farming alone is insufficient for bulk of the marginal farmers. The other activities such as dairying, poultry, sericulture, apiculture, fisheries etc. assume critical importance in supplementing their farm income. Livestock rising along with crop production is the traditional mixture of activities of the farmer all over the country, only the nature and extent various from region to region. It fits well with farm level infrastructure, small land base and abundant labour and ensures full utilization of by products.

Farming system approach addresses itself to each of the farmer enterprises; inter relationship among enterprises and between the farm and environment. Thus farming system research has the objective of increasing productivity of various enterprises in the farm while the crop production. But these approaches are not strictly compartment in the sense that any change in cropping system may bring about an inevitable change in the farming system.

Farming system approach introduces a change in farming technique for high production from a farm as whole with the integration of all the enterprises. The farm produce other than the economic products for which the crop is grown can be better utilized for productive purposes in the farming system approach. A judicious mix of cropping system with associated enterprises like dairy, poultry, piggery, fishery, sericulture etc. suited to the given agro-climatic conditions and socio economic status of farmers would bring prosperity to the farmer.

Classification of Farming Systems

Many farms have a general similarity in size, products sold and methods followed is called a type of farming or when farms are quite similar in kind and production of the crops and live stock that are produced and methods and practices used in production, the group is called as type of farming.

On the basis of the share of gross income received from different sources and comparative advantage, the farming systems may be classified as follows:

Classification of Farming Systems:

A) According to the Size of the Farm:

a) Collective farming.

b) Cultivation farming: i) small scale farming ii) large scale farming.

B) According to the Proportion of Land, Labour and Capital Investment:

a) Intensive cultivation.

b) Extensive cultivation.

C) According to the Value of Products or Income or on the basis of Comparative Advantages:

i) Specialized farming.

ii) Diversified farming.

iii) Mixed farming.

iv) Ranching.

v) Dry farming.

D) According to the Water Supply:

i) Rained farming.

ii) Irrigated farming.

E) According to:

I) Type of Rotation:

a) Lay system: i) Unregulated lay farming

ii) Regulated lay system.

b) Field system.

c) Perennial crop system.

II) Intensity of the Rotation:

a) Shifting cultivation.

b) Lay or fallow farming.

c) Permanent cultivation.

d) Multiple cropping.

F) Classification According to Degree of Commercialization:

a) Commercialized farming.

b) Partly commercialized farming.

c) Subsistence farming.

G) Classification According to Degree of Nomadic:

a) Total nomadic.

b) Semi nomadic.

c) Partial nomadic.

d) Transhumant.

e) Stationary animal husbandry.

H) Classification According to Implements Used for Cultivation:

a) Spade farming.

b) Hoe farming.

c) Mechanized or tractor farming.

Classification of Farming Systems- According to size of farm

According to the size of the farm:

1. Collective Farming:

It includes the direct collection of plant products from non-arable lands. It may include either regular or irregular harvesting of uncultivated plants; honeying and fishing usually go hand in hand with collection. Actual cultivation is not needed. The natural products like honey, gum, flower etc are collected. Such plant product may be collected from forestry area.

2. Cultivation Farming:

In this system, farming community cultivates the land for growing crops for obtaining maximum production per unit area.

i) Small Scale Farming: In this type, the farming is done on small size of holding and other factors of production are small in quantity and scale of production is also small.

Advantages:

i) Intensive cultivation is possible

ii) Labour problem do not affect the production.

iii) It is easy to manage the farm

iv) There is less loss due to natural calamities like frost, heavy rainfall, and diseases.

v) Per unit output increases

Disadvantages:

i) Small- scale farming cannot take advantages of various economic measures:

i) Cost of production per unit is more

ii) Mechanization of agriculture is not possible.

iii) Farmer does not get employment round the year.

ii) Large scale farming: When farming is done on large size holding with large amount of capital, large labour force, large organization and large risk is called large- scale farming.

In other words when the factors of production are large in quantity, the small of farming is said to be large. In India 40 to 50 hectares land holding may be said large scale farming but in countries like America, Canada and Australia even 100 ha farms are also called as small farms.

Advantages:

1. Production of large scale farming is more economical. The cost of production per unit is less.
2. Per unit production is increased.
3. Better marketing of agricultural products is possible. Processing, transportation, storage, packaging of produce is economical.
4. Costly machine like tractor, combined harvester can be maintained on the farm.
5. Subsidiary occupation such as dairy, poultry, bee keeping based on maintained on the farm.
6. Proper utilization of factors of production is possible.
7. Research work is possible.
8. It increases bargaining power of people.

Disadvantages:

1. If demand of produce decreases and production exceeds the market demand there will be more loss to the large farm.
2. In case of labour strike there will be more loss on the farm.
3. Due to natural calamities like frost, drought, flood, insects and diseases the large farm will suffer a lot.
4. It will be difficult to manage large scale farm.

Classification of Farming Systems

According to Production of Land, Labour and Capital Investment:

The farmer on given plot of land obtains more or less definite quantity of yield at any particular time. If he wants to increases his output he can either, i) Bring more land under cultivation, ii) Apply more labour and capital to the same piece of land.

a) Intensive Cultivation:

In intensive cultivation more labour and capital are used in the same piece of land. In other words land remains fixed in quantity while other factors are increased. If the same land is rare due to population pressure, while labour and capital are comparatively cheap, intensive cultivation is preferred than extensive cultivation. The application of intensive cultivation method depends mainly upon

i) Increasing population

ii) Technical improvement

In the earlier stages of development population was small and technical knowledge of agriculture was also limited, hence, extensive method was adopted but, as population increases intensive cultivation become necessary and improvement in technique makes its adoption possible.

b) Extensive Cultivation:

When more area is brought under cultivation to increase the output it is termed as extensive cultivation. In extensive cultivation land is available but availability of other factors increases less proportionately. A cultivator wishing to increases his output may follow either intensive method or extensive method but the selection of these two methods is based on cost. By following intensive cultivation can raise the additional output more cheaply, extensive method of cultivation will be useful. If on the other hand intensive cultivation seems to be the cheaper method he will naturally adopt it. If land is cheaper and it can be had at a normal cost while labour and capital are comparatively costlier, extensive cultivation will be cheaper method of obtaining increased output. In early times when land was plentiful extensive cultivation was followed. The extensive and intensive cultivation go side by side in a country for a certain period of time and afterwards intensive cultivation may become more important method. In most of the countries extensive and intensive methods of cultivation generally go hand in hand.

According to the Value of Products or Income or on the basis of Comparative Specialized Farming:

The farm in which 50% or more income of total crop production is derived from a single crop is called specialized farming or The farm in which only single crop is cultivated for selling in the market and the income of the farm depends mainly on that crop is called specialized farming by Hopkins.

According to the definition if 50% income is derived from paddy from any farm this is called paddy farm, similarly, sugarcane farm, wheat farm, vegetable farm, orchard farm etc.

Advantages and Disadvantages of Specialized Farming

Advantages:

1. Better use of land: More profitable to grow crops on land best suited to it. e.g. jute growing or cultivation on swampy land in West Bengal.

2. Better marketing: it allows grading, processing, storing, transporting and financing the produce.
3. Less equipment and labour.
4. Costly and efficient machinery can be kept: A wheat harvester, thresher can be maintained in a highly specialized wheat farm.
5. The efficiency and skill of the labor increase: Specialization allows a man to be more efficient and expert at doing a few things.
6. Farm records can be maintained easily.
7. Intensity of production leads to relatively large amount of output.
8. Better management: fewer enterprises on the farm are liable to be less neglected and sources of wastage can easily be detected.

Disadvantage

1. There is greater risk: When failure of crop and decreasing market price of the product, demand in market increases.
2. It is not possible to maintain soil fertility.
3. The productive resources i.e. land, labor and capital are not fully utilized.
4. Irregular income of the farm as they get income only once or twice in a year.
5. Proper Utilization of resources is not possible.
6. By product of crop are not property utilized, as livestock's are less in number.
7. Due to specialization of a single enterprise, the knowledge about other enterprises.
8. Does meet all the food needs of the family members of the farmer.

Diversified Farming and Its Advantages

A diversified farm is one that has several production enterprises or sources of income but no source of income equal as much as 50% of the total income from that source. On such farm farmers depends on several sources of incomes. It is also called as general farming.

Advantages

1. Better use of land, labour and capital: Better area land through adoption of crop rotations, steady employment of farm and family labour and more profitable use of equipment are obtained in diversified farming.
2. The farmer and labour remain engaged round the year in different activities.
3. Less risk to crop failure and market price of the product.
4. The byproducts of this farm can be utilize properly as cattle, poultry, birds, etc. are reared with crop production.
5. Regular and quicker return is obtained from various enterprises.
6. Soil fertility and erosion can be checked as land kept under cultivated throughout the year
7. Diversified farming is less risky than specialized farming.
8. Best use of all equipments.

Disadvantages:

1. Do not fetch desirable profit so long as co-operative marketing facility is not there.
2. Proper inspection of different enterprises is difficult.
3. It is not possible to farmer to maintain all types of machinery required for different crops.
4. Farm wastage in any farm is difficult to detect.

Mixed Farming and Its Advantages

Mixed farming

Mixed farming is one in which crop production is combined with the rearing of livestock. The live stock enterprises are complementary to crop production; so as to provide a balance and productive system of farming. In mixed farming at least 10% of its gross income must be contributed by livestock activity. The upper limit being 45% under Indian conditions. So, the farm on which at least 10 to 49% income is found from livestock is called mixed farm. In mixed farming cow and buffaloes are included with crop production. If farmers are rearing cows,

buffaloes, sheep goat, and fisheries with crop cultivation this type of farming is called diversified farming.

The scope of mixed farming to combination of crops and their complementary livestock enterprises of mixed farming would certainly include a vast majority of our farms, establishing a complementary relationship between crop and livestock enterprises.

Enterprises	**Contribution to gross income of farm**	**Farming type**
1. Cow and Buffalo only	10 to 49%	Mixed farming
2. Cow, buffalo and poultry	10 to 49%	Diversified farming

Advantages:

1. It offers highest return on farm business, as the byproducts of farm are properly utilized.
2. It provides work throughout year.
3. Efficient utilization of land, labour, equipment and other resources.
4. The crop by products such as straw, fodder etc. is used for feeding of livestock and in return they provide milk.
5. Manures available from livestock maintain soil fertility.
6. It helps in supplying all the food needs of the family members.
7. Intensive cultivation is possible.
8. If one source of income is lost he can maintain his family from other source of income.
9. Milk cattle's provide draft animals for crop production and rural transport.
9. Mixed farming increases social status of the farmer.

In India the livestock is highly connected with agriculture because animal power is the main source of power in agriculture. FYM is the main source for maintaining soil fertility and animals make good use of subsidiary and by-products

on farms and in turn they provide milk, under such circumstances mixed farming will most suit in Indian conditions.

Disadvantages:

1. Indigenous method of cultivation is used till now.
2. Draft and milch animals should be sold when they fail in production.
3. Healthy calf should be reared to replace age old animals.

Requirement of Mixed Farming:

1. Complicated management practices.
2. Sound cropping scheme.
3. Good cattle in suitable number.
4. Transport facility.
5. Marketing facilities.

Classification of Farming System

According to the Water Supply

i) Rainfed farming.

ii) Irrigated farming.

i) Rainfed Farming: Agriculture mainly depends on the rainfall in most part of the country. 80% of the total cultivated arable land is rain fed. Rain fed farming is very risky system of farming where the success of the crop depends on the cycle of the monsoon. Timely rainfall is the pre-requisite of this farming. The uneven rainfall is quite detrimental to crop production.

Characteristics of Rainfed Farming

1. Crop and varieties, which can withstand moisture stress, should be cultivated.
2. Kharif crops sown after receiving monsoon.
3. Not possible to adopt improved methods of cultivation only one or two crops are grown.
4. Crops rotation is not followed.

5. Soils of these areas are deficits in nutrient.
6. Mixed cropping should be practiced and adopt deep-rooted crops.
7. Short duration varieties fit well in rain fed areas.
8. The crops that are tolerant to drought should be cultivated.
9. Soil moisture should be preserved by mulching and FYM application.
10. Soil erosion which may be called “Creeping death” of the soil is a worldwide problem, so necessary measures should be adopted to keep the soil productive.

Principles of relevant components of environmentally sustainable farming systems should include.

1. Reduce soil erosion by improving soil conservations.
2. Inclusion of legumes and cover crops in crop rotations.
3. Agro-forestry as an alternate land use system.
4. Judicious use of organic waste.

ii) Irrigated Farming

The crop can be grown throughout the year; moisture is not a limiting factor.

Characteristics of integrated farming

1. Round the year cropping pattern is possible.
2. Intensive cropping is possible.
3. Production can be increased by proper utilization of productive resources.
4. Crop rotation can be executed properly due to adequate irrigation facility.
5. Manuring is safely done in irrigated crop.
6. The field experiment is possible, because of timely irrigation facility.

Classification According to Type of Rotation

a) Lay System

i) Unregulated lay farming.

ii) Regulated lay.

b) Field System.

c) Perennial Crop System.

I) Type of Rotation

The world rotation has two meanings according to the time period involved. There is long term alteration between various types of land use such as arable farming, tree farming, grassland use etc. in this rotation means the sequence of this basic type of land use on a given field. Within arable farming there is also the term crop rotation which means the short- term sequences of different arable crops on one field.

a) Lay System

In this system, several years of arable farming are followed by several years of grassed and legumes utilized for livestock production.

i) Unregulated Lay Farming: In this system natural vegetation grasses, bushy growth on pasture is allowed to grow during the period of fallow. This is an improved managed pasture.

ii) Regulated Lay System: During the period of fallow, certain types of grasses are grown or planted. These are the well managed pasture with fencing and adopting rotational grazing system.

b) Perennial Crop System: The crop which covers the land for many years e.g. tea, coffee, sugarcane. In some cases tree crops (oil palm, rubber) are alternated with fallow in other with arable farming, grazing etc.

II) Intensity of the Rotation

a) Shifting cultivation
b) Lay or fallow farming
c) Permanent cultivation
d) Multiple cropping

It is denoted by" R" simple and appropriate criteria for classification, which gives the true relationship between crop cultivation and following within the total length of one cycle of utilization.

$$R = \frac{\text{No. of years of cultivation}}{\text{Length of the cycle of land utilization}} \times 100$$

The length of the cycle is the sum of the number of years of arable farming. The number of fallow years "R" indicates the production of the area under cultivation in relation to a total area available for arable farming.

a) Shifting cultivation: R< 33%

In case there is more number of years of fallow than actual cultivation.

b) Lay or fallow farming: In this case R< 66% and 33%

c) Permanent cultivation; in this large area is cultivated and small area is left fallow. R> 66%.

d) Multiple cropping: where R = > 100%. If R= 150% means 50% area is under two crops in a year. If R = 300% means three crops in a year are being grown.

Classification According to Degree of Commercialization

Depending Upon the Produce Sold in the Market for Earning Money

a) Commercialized farming - More than 50% of the produce is for sale.
b) Partly commercialized farming - More than 50% of the value of produce is for home consumption.
c) Subsistence farming.

Virtually there is no sale of crop and animal products, but used for home consumption. Subsistence farming is a type of farming where the farmers of our country cultivate the crop on their land for the livings. Hence, the holding is small in size; so improved method of cultivation is not possible. They fail to meet the total requirement. They reared cattle, poultry, along with crop cultivation in limited land to meet their requirement.

Advantages

1. Utilizing productive resources profitably.
2. Farmers with their family members engaged throughout the year as they rearing cattle, poultry etc.
3. Farmers meet their demand from the income from cattle, poultry etc.
4. By product used properly.

Disadvantages

1. Fails to adopt improved crop cultivation technique due to small holding.
2. Cultivation mainly depends on monsoon rain.
3. Procurement of seed, fertilizer as and when required is difficult.
4. Income of this type of farm is very low.

Classification: According to Degree of Nomadic, Cropping pattern and Implements

1. Classification according to degree of nomadic

a) Total nomadic.

b) Semi nomadic.

c) Transhumant.

d) Partial nomadic.

e) Stationary animal husbandry.

Grassland Farming: In this type of farming, it involves the rearing of animal for economic production and is classified on the basis of nomadic.

a) Total nomadic: In this system, the animal owners do not have permanent place of residence. They do not practice regular cultivation. Their families move with the herds.

b) Semi nomadic: Animal owners have a permanent place of residence, while supplementary cultivation is practiced. However, for long periods of time, they travel their herd to distant grazing areas.

c) Transhumant: Means seasonal migration of livestock to suitable grazing ground or it is the situation in which farmer with a permanent residence sends their herd with herdsman for long period of time to distant grazing areas.

d) Partial nomadic: Farmer has permanent residence and who have herds at their disposal, which remains in the vicinity.

e) Stationary animal husbandry: Where the animals remain on the holding or in the village throughout the entire year.

2. Classification According to Cropping Pattern and Animal Activities:

In this system classification according to the leading crops and livestock activities of the holdings.

e.g**.** Paddy-wheat holding, coffee - banana holding. Rice-Jute holding, Sugarcane farming.

3. Classification According to Implements Used for Cultivation

a) Spade farming: Manual labour is used.

b) Hoe farming: To use bullock power for cultivation.

c) Mechanized or tractor farming: Power operated implements are used for cultivation.

eg. plough, tractor.

Characteristics of Indian Agriculture:

1. Uneven distribution of land.
2. Fragmentation of holding.
3. Existence of small and marginal farmers.
4. Regional variation.
5. Dependence of seasonal rainfall.
6. Production of different kind of crops.
7. Low productivity of land.
8. Increasing of disguised unemployment.
9. Disorder in marketing of Agricultural products.
10. Weak land reformation.

Chapter 2. Historical developments of Farming Systems in India under different situations

India with 2.2 per cent of global geographical area supports more than 15 per cent of the total world population, 70 per cent of whom depend on agriculture. It also supports nearly 15 per cent of the total livestock population of the world. One-third of the gross national product comes from agricultural sector. By 2050 A.D., 349 MT of food, 25 MT of vegetable oil and 92 mm^3 of industrial wood shall be needed for approximately 1667 million people. As of now, out 328.73 M ha of geographical area approximately 18 per cent is under forest; only 13.5 per cent is not available for cultivation. Total problem areas constitute 173.65 M ha which include areas subject to wind and water erosion (145 M ha), water-logged areas (8.53 M ha), alkali soils (3.58 M ha), saline and coastal sandy areas (5.50 M ha), ravines and gullies (3.97 M ha), shifting cultivation (4.91 M ha) and reverie torrents (2.73 M ha). Besides 40 M ha are prone to flood and 260 M ha are drought prone. Thus, the net sown area is 136.18 M ha (41.42 per cent of the total geographical area).

Unlike industries, agriculture is practiced by 105 million farm families who live in 0.6 million villages. More than 40 per cent of them are below the poverty line. Nearly 85 million farm families belong to small and marginal categories. In spite of increase in food production, after the independence of the country, only in north-western India, per capita food production has increased and it has declined in other parts of the country. The per capita availability of land during the same period has declined from 0.48–0.15 ha by 2000 A.D. Per capita investment in agricultural infrastructure is the lowest in eastern India, where the density of population is the highest. Only 25–30 per cent of the modern agricultural technologies have reached the farmers. This is often because the technology has not been consistent with conditions of the farm situations.

The benefits of modern technology have, however, been restricted to favourable farming situations. Only 44 out of 453 districts are contributing half of the total food grain basket of the country. This clearly suggests that the technology supposed to be scale and resource neutral has been confined to the districts with

favourable farming conditions. Since there is no further scope for horizontal expansion of land for cultivation, the only alternative left is for vertical expansion and diversification of farm enterprises with less demand on space and time particularly for small and marginal farmers (constituting 76 per cent of the farming community) who do not have much of resources, especially in rainfed areas (70 per cent of the total cultivated land). The new farming system research strategy should, therefore, necessarily concentrate on developing technology with participatory approach within the biophysical and socioeconomic environments in which the farmers operate.

Present Research Thrust and its Limitations

The existing programme of the ICAR is a component research and there is no system-based programme. The response of a component in isolation does not necessarily fit into a systems perspective. The individual programme ignores the socio-economic and biophysical aspects of the farming community.

The limitations of the conventional model of agricultural research and extension in dealing with interactive matrix are well recognized. The reductionist component research approach curtails system perspective resulting in wide gap in technology development and utilization particularly for the resource poor farmers. Systems research takes into consideration appropriate perspective in handling a complex undertaking such as farm enterprises, diversification and their interaction on farm productivity. The challenge is to design practical integrated farming systems that can be adapted to regions, minimize energy and base inputs, and are sustainable. Design and implementation of integrated farming systems require collaboration among agricultural, social, economic, and ecological disciplines. They also depend upon the full participation of farmer. It may be probably impossible to develop efficient integrated farming systems without interdisciplinary planning and implementation. A major challenge to agricultural research and

development is the need for modifying institutional structures and research and development methodologies to allow such collaboration.

Historical Development

World population has been increasing by leaps and bounds. India's population is expected to reach 1370 and 1660 m. in 2030 and 2050 A.D. respectively. The country's food production has reached an all time high of 204 MT during 2000. Food production level of 289 and 349 MT will be needed to satisfy the projected population in 2030 and 2050 A.D. respectively. The net cultivable area is 142.8 M ha. Unlike the population spurt and corresponding food need for 2050, there is every chance that the land area under cultivation will decrease due to diversion of some of the cultivable area to buildings and industrial purposes. It is anticipated that the land area available for cultivation in 2050 would be 137 M ha. So, it becomes necessary to increase the productivity almost to double of what we are producing today. This could be made possible by putting the land, both irrigated and rainfed, under intensive cultivation. Fortunately most of our country lies in tropics and so is blessed with abundant solar energy, thus making cropping round the year possible. The only way to increase agricultural production is to increase the productivity per unit area per unit time. The cropping systems, genotypes, geometry of planting and management practices are designed to increase the productivity, simultaneously making efficient use of available resources and stabilising yields.

Modern agriculture emphasises two dimension *viz*., time and space. Time concept relates to increasing crop intensification in a situation where there is no constraint for inputs including irrigation. In other words increasing the cropping intensity in areas where the production potential *viz*., land is under-utilized even with full resource potential. It is a time bound program where, for most of the field crops, it is considered for 365 days or 12 months. In the case of long duration and perennial crops the duration of each rotation will vary from two or three years depending upon the duration of the constituent crops. The areas, where only one

crop (100 per cent) two crops (200 per cent) three crops (300 per cent), are raised in an year, leaving the land fallow for two to eight months the cropping intensity has to be increased to 200, 300 and 400 per cent respectively within one year. This will ultimately help to enhance the productivity. In case of rainfed areas where there is no possibility of increasing the intensity of cropping, the other concept *viz*., 'space concept' can be applied. By raising of crops/other agricultural allied components in a vertical dimension, the land equivalent area can be increased. Thus by making use of these time and space dimensions either in irrigated or in rainfed areas within specified time span, say a year, and unit area of land, a hectare, productivity is sought to be increased by repeated and/or intensified cropping. This calls for urgent action on the part of Agronomists and Scientists of related disciplines to devote their attention on the design and testing of cropping systems for different regions. Income through arable farming alone is insufficient for bulk of the marginal farmers. Activities such as dairying, poultry, fish culture, sericulture, biogas production, mushroom cultivation, agro forestry etc. assume critical importance in supplementing their farm income. These activities fit well with farm level infrastructure and ensure full utilisation of by-products.

Integrated Farming System (IFS) is the answer to increase food production and farm income and for improving nutrition of the small-scale farmers with limited resources without any adverse effect on environment and agro-ecosystem. In a cropping system, the amount of by-products can be as high as or higher than marketable produce. This may go to waste if not utilised in an animal enterprise. Hence, integration of different agricultural allied enterprises with crop activity as base will provide a way to recycle low cost produces at farm level/no cost waste materials of farm from one enterprises to another and thus reduce the cost of production of the economic produce and finally to enhance the net income of the farm as a whole.

Chapter 3. Farming System Concept and components, interaction between components

Sustainability is the objective utilization of inputs without impairing the quality of environment with which it interacts. Therefore, it is clear that farming system is a process in which sustainability of production is the objective.

The overall objective is to evolve technically feasible and economically viable farming system models by integrating cropping with allied enterprises for irrigated, rained, hilly and coastal areas with a view to generate income and employment from the farm.

The Specific Objectives are:

1. To identify existing farming systems in specific areas and access their relative viability.
2. To formulate farming system model involving main and allied enterprises for different farming situations.
3. To ensure optional utilization and conservation of available resources and effective recycling of farm residues within system.
4. To maintain sustainable production system without damaging resources/ environment.
5. To rise over all profitability of farmhouse hold by complementing main/allied enterprises with other.

Scope of Farming System:

Farming enterprises include crop, livestock, poultry, fish, bee keeping, sericulture etc. A combination of one or more enterprises with cropping when carefully chosen, planned and executed gives greater dividends than a single enterprise, especially for small and marginal farmers. Farm as a unit is to be considered and planned for effective integration of the enterprises to be combined with crop production activity.

Integration of Farm Enterprises Depends on Any Factors Such as:

1. Soil and climatic features of the selected area.
2. Availability of the resources, land, labor & Capital.

3. Present level of utilization of resources.
4. Economics of proposed integrated farming system.
5. Managerial skill of farmer.

System Approach

Management practices are developed for individual crops and recommendations are made for individual crops. The residual effects of individual crops are not considered in crop-based recommendation; in this resources are not utilized efficiently. To a farmer, instead of crop, land is a unit and management practices should be for all crops that are to be grown on a piece of land. Therefore, system approach is applied to agriculture for efficient utilization of all resources, maintaining stability in production and obtaining higher net returns.

Cropping System

Cropping system, an important component of a farming system, represents a cropping pattern (i.e. the proportion of area under various crops at a point of time in a unite area) used on a farm and their interaction with farm resources, other farm enterprises and available technology, which determine their makeup.

It is defined, as the order in which the crops are cultivated on a piece of land over a fixed period or cropping system is the way in which different crops are grown. In the cropping systems, sometimes a number of crops are grown together or they are grown separately at short intervals in the same field.

Classifications of Cropping System

Depending on the resources and technology available, different types of cropping systems are adopted on farms, which are as below

Mono-cropping or Single Cropping: Mono-cropping refers to growing only one crop on a particular land year after year or Practice of growing only one crop in a piece of land year after year e.g. growing only crops in dry lands or only said crops

in diary lands (Lands situated in river basins which often remain flooded during rainy season). This is due to climatologically and socio-economic conditions or due to specialization of a farmer in growing a particular crop.

Groundnut or cotton or sorghum is grown due to limitation of rainfall. Flue-cured tobacco is grown in Günter (A.P.) due to specialization of a farmer in growing a particular crop. Rice crop is grown, as it is not possible to grow any other crops, in canal irrigated areas, and under water logged conditions.

Monoculture: Practice of repetitive growing only one crop irrespective of its intensity as rice-rice-rice in Kerala, West Bengal and Orissa.

Sole Cropping: One crop variety grown alone in pure stand at normal density.

Multiple Cropping or Polycropping: It is a cropping system where two or three crops are gown annually on the same piece of land using high input without affecting basic fertility of the soil.

Growing two or more crops on the same piece of land in one calendar year is known as multiple cropping. It is the intensification of cropping in time and space dimensions i.e. more number of crops within a year and more number of crops on the same piece of land at any given period. It includes inter-cropping, mixed cropping and sequence cropping.

Multiple cropping is a philosophy of maximum crop production per acre of land with minimum of soil deterioration e.g.

1) Rice-potato-green gram.
2) Rice-mustard-maize.
3) Rice-potato-sesame.
4) Jut-rice-potato.

Cropping intensity is more that 200 per cent when the farm as a whole is considered; the Multiple Cropping Index (MCI) is determined by the number of

crops and total area planted divided by the total arable area. When the value is three or more, it is said to be most promising farm. This is also called as intensive cropping.

1. Polyculture: Cultivation of more than two types of crops grown together on a piece of land in a crop season.

e.g.

1) Subabul + Papaya + Pigeon pea + Dinanath grass.

2) Mango + Pine apple + Turmeric

3) Banana + Marigold + berseem.

Relay Cropping: Growing the succeeding crop when previous crop attains its maturity stage-or-sowing of the next crop immediately after the harvest of the standing crops. Or it is a system of cropping where one crop hands over land to the crop in quick succession.

e.g.

1) Paddy-lathyres

2) Paddy-Lucerne.

3) Cotton-Berseem.

4) Rice-Cauliflower-Onion-summer gourds.

Overlapping Cropping: In this system, the succeeding crop is sown in the standing crop before harvesting. Thus, in this system, one crop is sown before the harvesting of preceding crops. Here the lucre and berseem are broadcasted in standing paddy crop just before they are ready for harvesting.

Advantages:

1. Minimum tillage is needed for relay cropping and primary cost of cultivation is less.

2. Weed infestation is less, as land is engaged with crops year round.

3. Crop residues are added in the soil and thus more organic matter.

4. Residual fertilizer of previous crops benefits succeeding crops.

Components of Farming Systems

In the integrated farming system, it is always emphasized to combine cropping with other enterprises, activities. Many enterprises are available and these include cattle maintenance sheep or goat rearing, poultry, piggery, rabbit rearing, bee keeping etc. Any one or more can be combined with the cropping.

Cattle Maintenance:

1) Draft breeds,
2) Dairy breeds,
3) Dual purpose,
4) Exotic breeds.

1. Buffaloes: Important dairy breeds of buffalo are murrah, mehsana, zefarabadi,Godavari.

Feeding: Cattle feed generally contains fibrous coarse low nutrient straw material. Roughage is basic for cattle ration and includes legumes non- legume hays, straw and silage of legume and grasses. Per day requirement @ 1 kg concentrate per 2 lit of milk, green fodder (20- 30 kg), straw 5-7 kg & water – 32 lit.

2. Sheep Rearing: Sheep are well adapted to many areas. They are excellent cleaners and make use of much of waste feed. Consume roughage, converting a relative cheap food into a good cash product. Housing not expensive.

Feeding: 1-2 kg of leguminous hay per day. Protein supplied through concentrate as groundnut cake.

3. Goat Rearing: In India, activity of goat rearing under different environments. The activity is also associated with different systems such as crop or animal based, single animal or mixed herd small or large scale. Goat is mainly reared for meat, milk hide and skin meat preferred in India, A goat on hoof fetches a better price than a sheep on hoof.

Feeding: Per head nutrients requirement to goat is low. Hence, they are suitable for resources poor small farmers with marginal grazing lands they eat plants and leaves

of tree, which any other animals not touch. Goat eats 4- 5 times that of body weight concentrate of maize, groundnut cake etc. and clean and fresh water.

4. Poultry: Poultry is one of the fastest growing food industries in the world. Poultry meat accounts for about 27% of total meat consumed worldwide poultry industry in India is relatively a new agricultural industry. Egg production may reach up to 5000 crores and broiler meat production 330 thousand tones (by 2000) the average global consumption is 120 eggs per person/ year and in India it is only 32- 33 eggs per capita/ year. To meet the nutritional requirement the per capita consumption estimated at 180 eggs 9 kg meat/ year.

Feed: The feed conservation efficiency of the bird is superior to other animals. About 60 – 70 % of the total expenditure on poultry farming is spent on the poultry feed. Hence, use of cheap and efficient ration will give maximum profit cereals- maize, barley, oats, wheat, rice – broken mineral/ salt – limestone, salt manganese.

5. Duck rearing: Ducks account for about 7 % of the poultry population in India. They are popular in cereal and logged states like West Bengal. Orissa, A.P, T. Nadu, they have production potential of about 130- 140 eggs/ bird/ year. These can rear in marshy riverside Westland. Duck farming can be a better alternative.

Feeding: Eating fallen grains in harvested paddy fields, small fishes and other aquatic materials. A variety of crop residues and insects in the farm are also available.

6. Turkey rearing: Turkey is a robust bird and can be reared in humid tropics. It actively feed on a variety of crop residues and insects in the farm.

7. Piggery: Pigs are maintained for production of pork.

8. Rabbit Rearing: In India is of recent origin though hunting of wild rabbits for meat is not uncommon. Rabbit can be easily reared with relatively less concentrate feed with high production rate.

9. Bee Keeping: Bee keeping is one of the most important agro- based industries which do not required any raw material like other industries. Nectar and pollen from flower are the raw material, which available in plenty in nature. Bee keeping can be started with a single colony.

Honey collection: Honey should have good quality. Qualities such as aroma, color, consistency and floral sources are important. Honey is an excellent energy food with an average of about 3500 calories per kg it is directly absorbed into the human blood stream requiring no digestion.

10. Aquaculture: Ponds serve as domestic requirement of water, supplementary irrigation to crop and pisciculture with the traditional management, farmer obtain hardly 300 – 400 of wild and culture fish/ ha/ year. However, polyfish culture with the stocking density of 7500 fingerlings and supplementary feeding will boost the total biomass production.

Species: cattle, rohu, common carp, silver carp, and grass carp (feed on aquatic plants).

Management: Pond depth – 1.5- 2.0 m, water should be slightly alkaline, pH- 7.5- 8.5. If the pH less than 6.5, it can be adjusted with addition of lime, higher pH (> 8.5) can be reduced with addition of gypsum. Application of fresh dung may also reduce high pH in the water.

The fish are to be nourished with supplementary feeding with rice bran and oilseed cakes. This will enable faster growth and better yield. Each variety of crops stocked to 500 fingerlings with the total of 5000- 8000/ ha. This gives 2000 to 5000 kg/ha of fish annually.

11. Sericulture: Keeping of silk moths and their larvae for the production of silk or sericulture is defined as a practice of combining mulberry cultivation; silkworm rearing and silk reeling. Sericulture is a recognized practice in India. The total area under mulberry is 240 thousand ha in the country. It plays an important role in

socio- economic development of rural poor in some areas. Climatic condition favorable for mulberry and rearing of silk worms throughout the year. Karnataka is the major silk producing state in India (temp 21 to 30° C), in Kashmir climate suit from May to October.

Moriculture: Cultivation of mulberry plants is called as moriculture. The crop yield well for 12 years. Yield of mulberry leaves is 30- 40 t/ ha/ year.

Rearing: Eggs are allowed to be laid over a cardboard. In Bamboo tray rice husk is spread. Tender chopped mulberry leaves are added to the tray. The hatched out larvae are transferred to the leaves it is important to change the leaves every 2 – 3 hours during the first 2 – 3 days. The cocoon constructed with silk. The cocoons required for further rearing are kept separately and moths are allowed to emerge from them.

12. Mushroom Cultivation: Mushroom is an edible fungi great diversity in shape, size and colour. Essentially mushroom is a vegetable that is cultivated in protected farms in a highly sanitized atmosphere; mushroom contains 90 % moisture with in quality protein, fairly good source of vitamin C and B complex. It is rich source of mineral like Ca, P, K & Cu. They contain less of fat and CHO and are considered good for diabetic and blood pressure patients.

Varieties: 1) Oyster mushroom 2) Paddy straw mushroom- Volvarilla volvacea 3) White button mushroom- Agaricus gisporus (var, A-11, Horst V3).

13. Biogas Plant: Biogas is a clean, unpolluted and cheap source of energy, which can be obtained by a simple mechanism and little investment. The gas is generated from the cow dung during anaerobic decomposition. Biogas generation is a complex bio- chemical process, celluloytic material are broken down in methane and CO_2 by different group of micro- organism. It can be used for cooking purpose, buring lamps, etc. Biogas near to kitchen & cattle shed to reduce cost of gas transfer and cow dung transport, sunlight is important for temperature.

Biogas slurry: Slurry is obtained after the production of biogas. It is enriched manure; another positive aspect of this manure is that even after weeks of exposure to the atmosphere the slurry does not attract fleas and worms. Dry slurry contains about 1.8 % N, 1.10 %P & 1.50% K.

Chapter 4. Cropping System, complementary and competitive interaction

Cropping System Related terminology

Cropping system: A cropping system refers to the principles and practices of cropping and their interaction with farm resources, technology, aerial and edaphic

environment to suit the regional or national or global needs and production strategy. It is an important component of farming system.

Cropping pattern: The yearly sequence and spatial arrangement of crop or of crops and fallow on a given area (a farm), region, province or country apportioning due consideration to natural features (soil and climate), crop efficiency, and capability, socio-economic structure, technological and extension infra-structure (changeable) and the national agricultural policy.

Multiple cropping

Multiple cropping refers to intensification of cropping both in time and space. It includes sequential cropping, inter-cropping and mixed cropping.

a) Sequential cropping: Growing two or more crops in a sequence on the same field in a farming year (twelve months) for irrigated land and is limited to the period of adequate soil moisture availability for crop growth in semi-arid & arid areas. The succeeding crop is planted after the preceding crop has been harvested. Crop intensification is only in time dimension. There is no inter-crop competition. Farmers manage only one crop at a time in the same field.

b) Intercropping: It refers to growing of two or more dissimilar crops simultaneously on the same piece of land, base crop necessarily in distinct row arrangement. The recommended optimum plant population of the base crop is suitably combined with appropriate additional plant density of the associated/component crop. The objective is intensification of cropping both in time and space dimensions and to raise productivity per unit area and inputs by increasing the pressure of plant population. The following four types of inter-cropping are identified.

i) Mixed inter-cropping: Growing component crops simultaneously with no distinct row arrangement. This is commonly used in labour intensive subsistence farming situations.

ii) Row inter-cropping: Growing component crops simultaneously in different row arrangement. This is used in mechanized agriculture.

iii) Strip inter-cropping: Growing component crops in different strips wide enough to permit independent cultivation but narrow enough to the crop to interact agronomically.

iv) Relay inter-cropping: Growing component crops in relay, so that growth cycles overlap. It necessarily does not mean planting of succeeding crop before flowering stage of preceding crop or attainment of reproductive stage of preceding crop. It refers to planting of succeeding crop before the harvest of preceding crop, planting of succeeding crop may be done before or after flowering before or after attainment of reproductive stage, completion of active life cycle, senescence of leaves or attainment of physiological maturity.

Mixed cropping: Mixed cropping is growing of two or more crops simultaneously on the same piece of land seeded either after the seeds of the crops intended to be grown mixed or sowing alternate rows in various replacement ratios. This may or may not have distinct row arrangement and the mixed plant community faces inter and intra row competition with a different plant type/variety. The basic objective in mixed cropping is minimization of risk and insurance against crop failure due to aberrant weather conditions. In inter-cropping systems, pressure of plant density per unit are is more than that in a sole cropping system, while in mixed cropping the plant population pressure is generally equal to sole cropping and in some cases it may even be less than sole cropping system. Besides the above few other terms commonly used in cropping systems approach are defined below:

Monoculture: The repetitive growing of the sole crop on the same piece of land. It may either be due to climatological limitation or due to specialization by a farmer to grow a particular crop.

Staggered planting: It means sowing of a crop is spread over and around optimum period of planting either to minimize risks or to use labour & machinery more

effectively or to minimize competition (in inter-cropping) or to prolong the period of supply to the market or the factories.

Ratoon cropping: The cultivation of crops re-growth coming out of roots or stalks of the preceding crop after harvest, although not necessarily for grain is termed as ratoon cropping/ratooning.

Mixed farming: It is defined as a system of farming on a particular farm (regardless of size) which includes crop production, raising of livestock, poultry, fish and bee keeping, and/or tress to sustain and satisfy as many necessities of the owner (farmer) as is possible. Subsistence is the objective here. It is based on the principle of give and take. Farm animals feed on farm produce and in return manure is given to the crops.

Cropping Systems:

Efficient cropping systems for a particular farm depend on farm resources, farm enterprises and farm technology because farm is an organized economical unit. The farm resources include land, labour, water, capital and infrastructure. When land is limited intensive cropping is adapted to fully utilized available water and labour when sufficient and cheap labour is available, vegetable crops are also included in the cropping systems as they required more labour. Capital intensive crop like sugarcane, banana, turmeric etc. find a space in the cropping system when capital is not a constraint. In low rainfall regions (750 mm/annum) mono cropping is followed and when rainfall is more than 750 mm, intercropping is practiced, with sufficient irrigation water, triple and quadruple cropping is adopted, when other climatic factors are not limiting farm enterprise like daring, poultry etc. also influenced the type of cropping system. When the farm enterprises include dairy, cropping system should contain fodder crops as components change in cropping system take place with the developments of technology. The feasibility of growing for crop sequences in Genetic alluvial plains inputs to multiple cropping.

Plant Interactions and Its Types

The interaction between different components crops: In intensive cropping, when crops are grown in association (intercropping) or sequence (sequential cropping) interaction between different component crop species occurs, which is essentially a response of one species to the environment as modified by the presence of another species (commonly referred to as interference or interaction).

Interference can be divided into two

a) Removal reactions of one plant on its environment and

b) Additive reactions when something is added.

When some factor is removed from the environment, the resulting response of neighboring species can be negative, positive or neutral. Competition among plants is one example for removal interactions. Some such additive interactions are allelopathic and symbiosis. When crops are grown in sequence, residual effect of the preceding crop influences the succeeding crop this may be harmful or helpful. The toxic chemicals (allelopathic chemicals) left in the soil by the roots of sunflower crop inhabit germination of the succeeding crop. The stubbles of sorghum with high C: N ration causes immobility of nitrogen, thus causing nitrogen deficiency in early stages of the succeeding crops. The roots of legume crops and their residues add nitrogen to soil.

The interaction may be:

1. Competitive
2. Non-competitive and
3. Complementary.

1. Competitive Interaction: - One species may have greater ability to use the limiting factor and will gain at the expense of the other and this is called as competitive interaction or interference. Or when one or more growth factors are limiting, the species that is better equipped to use the limiting factors(s) will gain at the expense of the other and this is called as competitive interaction. In mixed crop communities, if the associated species are to share their growth from a limited pool

of recourses such as light water or nutrients, then it is non-competitive interaction or interference.

2. Non-competitive:- If the crops are grown in association and the growth of either of the concerned species is not affected, such type of interaction is called non-competitive interaction or interference. Or if these resources (growth factors) are present in adequate quantities as a result of which the growth of either of the concerned species is not affected, then it is non-competitive interaction or interference.

3. Complementary: - If one species is able to help the other it is known as complementary interaction. Or if the component species are able to exploit to supply of growth factors in different ways (temporal or spatial) or if one species is able to help the other in supply of factor (like legumes supplying part of N fixed by symbiosis to non-legumes), it is complementary interaction or interference. This also referred to as Annidation.

Interaction in Inter-Cropping: Competition between the associated crops in mixed crop communities has been discussed by Willey (1979).

Competition for Solar Radiation:

Solar radiation falling above 700nm is called the near infra red radiation (NIR). PAR is commonly known as light and s directly related to photosynthetic rates of component crop while NIR is the primarily energy source deriving evapotranspiration, sensible heat exchange and other photomorphogenic process. These two types of radiation considered together under this section.

Beer's law describes the light penetration into a crop canopy if the foliage distribution is uniform in horizontal line.

$I = Ice – KL$

Where I = light flux density, to a horizontal surface below L units of LAI,

Io is the light flux density to a similar surface above canopy.

e- Base of natural logarithm.

k- Is the extinction coefficient.

In sparse plant stands or with low LAI beers low under estimate light interception.PAR is directly related to photosynthetic rate of component crops. Intercropping can increase light interception by as much as 30-40%. The taller crop in the intercropping systems intercepts most of the solar radiation while shorter component suffers. In some intercropping systems, both crops utilize solar radiation efficiently. In Ground nut red gram intercropping system, light interception is prolonged as red gram starts growing after harvesting of groundnut. If the component crops have different growth durations the peak demand for light occurs at different times. In maize + green grams intercropping system, green gram flowers in 35 days after sowing and is harvested 65 days after sowing, peak light demand for maize occurs at 60 days after sowing when green gram is ready for harvest. In such intercrops, there is less competition among component crops and higher solar radiation is intercepted in intercropping system than in pure crops.

Proper choice of crops and varieties, adjustment of planting density and pattern are the techniques to reduce competition and increase the light use efficiency.

1. When one component is taller than the other in an intercropping system, the taller component intercepts greater share of light. As a result, the growth rates of the two components will be proportional to the quantity of the two component will be proportional to the quantity of PAR they intercept provided other growth factors are no limiting and the crops are in their vegetative.

2. The inclination of leaves greatly influences the amount of light intercepted by the taller component and the amount that is available to shorter components. For example, one unit of LAI of prostrate-leaved perennial rye grass (*Trifolium ripens*) absorbed 50% of the incoming light where as the some LAI of erect leaved perennial rye grass (*Lolium pereme*) absorbed only 26%.

3. Light interception in a sorghum (*Sorghum bicolor*) based intercropping system was studied by various workers. Light was measured influx units from 45th-90th day after sowing sorghum at 15 day interval. Light interception was expressed as percentages of light on the top of the canopy of each crop.

The redgram/sorghum intercropping system makes better use of growth resources, particularly light, was brought out by Willey *et al*. (1981). Ideally the taller component should have more erect leaves and the shorter component more horizontal leaves. If these to be planted in alternate rows, there would be less competition for light. Under intercropping situations, the component crops are grown in such a way that competition for light is minimized; this can be achieved by proper choice of crops and genotypes, the shorter components being harvested sufficiently early so that the later harvested component is not greatly affected.

Effect of solar radiation on evapotranspiration, sensible heat flux and photomorphogenic effect has not been investigated in detail in mixed crop communities. evapotranspiration would be much less for the lower canopy in intercropping, leading to less water stress. Finally, the success of the intercropping system would be very much dependent on the photoperiod adaptability of the component crops.

Competition of Other Interactions

Other Interactions:

Allelopathic is any direct- or – indirect harmful effect that one plant has on another through the relates of chemical substances- or – toxins into the root environment. Some crops may be unsuitable to be grown as inter- crops because they may produce and excrete toxins into the soil, which are harmful to other components. Allelochemical produced from the leaves of eucalyptus globules drastically reduce the germination of mustard sown underneath. On the contrary, negative allelopathic i.e stimulates the growth of this associated crop by release hormone like substances, is also possible. But release of N from root nodules of

legume is not considered to be a form an allelopathic. The chemical released by one species may inhibit species of plants other than the one releasing it (allow inhibition) or may inhibit more strongly plants of the producer species itself (auto inhibition). Toxic substances may be converted into active substances by some micro- organisms (functional allelopathic). The type quantity of allelochemical produced will vary depending on the environment and genetic makeup of the plant.

Some allelochemical may be produced by the aerial portion of the plant and may reach the ground through raindrops, falling leaves or insects, inhibition the growth of the species growing underneath. Allelochemical produced from the leaves of Eucalyptus globules drastically reduced the germination of mustard (*Brassica spp*.) seed sown underneath. Many plants exude organic substances from their roots and some of these roots exudates act as allelochemicals inhibiting the growth of the neighboring species living roots of walnut (*Juggles nigra*) cucumber (*Curcumas sativa*) and path (*Prunes persia*) are known to exude toxic substances which inhibit the growth of the plants growing near them.

Annidations: Annidations is complementary use of resources by exploiting the environmental supplies in different ways by the component of a community or it refers to complementary interaction which occurs both in space and time.

a) Annidations in Space: The leaf canopies of component crops may occupy different vertical layers with taller component tolerant to strong light and high evaporative demand and charter component favoring shade and high relative humidity. Thus, one component crop helps the outer. Multistoried cropping in coconut and planting of shade trees in coffee, tea and cocoa plantation use this principle. Similarly, root system of component crops exploits nutrients from different layers of soil and thus utilizing the resources efficiently. Generally, one component with shallow root system and another with deep root system are selected for intercropping as in set aria (shallow red gram (deep) intercropping system.

b) Annidations in Time: When two crops of widely varying duration are planted, their peak demands for light and nutrients are likely to occur at different periods, thus reducing competition. When the early maturing crops are harvested, condition becomes factorable for the late maturing crops to put forth its full vigor. This has been observed in sorghum + red gram, groundnut + red gram and maize + green gram intercropping system.

c) Other Complementary Effect in Intercropping Systems: In an intercropping system, involving as legume and non-legume, part of the nitrogen fixed in the root nodule of the legume may become available to the non-legume component. The numerous reports of such beneficial effect of legumes on non-legumes are available in literature. The presence of rhizosphere micro flora and mycorrhiza, one species may lead to mobilization and greater availability of nutrients not only to the species concentrated, but also to the associated species. Another example is the provision of physical support by one species to the intercropped is climbing species may improve the yield of the climber. e.g. coconut + pepper and maize + beans. The taller component acts as wind barrier protecting the short crop as in maize+ groundnut, onion + castor, turmeric + castor.

Chapter 5. Effect of preceding crops and associated crops

Effect of preceding crops on the following crops

In sequential cropping, sole crops are sown in sequence and so competition for soil factors is not a problem. But management of water and nutrients is modified by the effect of the preceding crop on the succeeding crop. This includes the complementary effects such as release of N from the residues of the previous crop, particularly legumes, to the following crops and carry-over effects of fertilizers applied to the preceding crops and adverse effects such as allelopathy, temporary immobilisation of N due to wide C/N ratio of the residues and carryover of pests and diseases.

Cotton raised after legumes performed better than when it followed cereals, particularly sorghum. Yield declined considerably in cotton-cotton rotations. Raising *bhendi, a* crop belonging to *Malvaceae,* to which cotton also belongs, also reduced the yield of the following cotton. Sowing cotton in *ragi* stubbles improved the yield and N uptake of cotton.

Sorghum had depressing effects of the following crops, especially if they were cereals. Sunflower was a good preceding crop for *ragi,* when the time allowed between the two crops was sufficiently long for the decomposition of sunflower residues (at least a fortnight).

Continuous monoculture of rice resulted in gradual reduction in grain yield but this yield reduction could be overcome by good management, particularly adequate fertilization. Introduction of pulses as summer crops resulted in improved yield of rice in *kharif*. Raising cotton and greengram in summer increased the yield of the following rice but cereals, particularly sorghum and maize, as preceding crops had adverse effect on rice yield.

Reviewing the rotation experiments conducted in Senegal, observed the following:

1. Groundnut grows well after any crop but benefits little from the previous crop. Groundnut is a good crop to precede any cereal, especially rainfed rice.
2. Cotton yields better after maize, groundnut and pearl millet but is affected by preceding sorghum. Successive cotton is not good. Maize yields well after cotton.
3. Any crop following rainfed rice yields poorly. Successive rainfed rice is not good.
4. Sorghum grows well after cotton and groundnut. In light soils, any crop following sorghum is affected. Cotton and groundnut are good preceding crops for sorghum. Sorghum after sorghum is not good.

It is evident from the above discussion that continuous culture of the same crop causes drastic yield reductions in course of time partly because of pest build-up and partly because the same layer of soil is repeatedly foraged and depleted. Though adequate fertilization may offset the yield reduction to start with, it is not a good

proposition in the long run. Even culture of crops that act as alternate host of pests will result in similar yield reduction (e.g. Cotton-bhendi).

Any fast growing cereal would deplete the soil and leave behind stubbles and residues with wide C/N ratio, which take a long time to decompose. During the process of decomposition, part of the soil N is immobilised affecting the following crop. Sorghum is a good example and sorghum effect noticed in low fertility soils is due to temporary immobilisation of N. *Ragi* seems to be an exception, mainly because the residues left behind decompose rapidly resulting in mineralisation of soil N.

Though it has been reported that sunflower residues produce allelo-chemicals during decomposition, if sufficient time (a fortnight) is allowed to lapse between the harvest of sunflower and sowing of the subsequent crop, the allelochemicals produced disappear and the rapidly decomposing residues cause mineralisation of soil N, benefitting the subsequent crop.

Cotton feeds in the deeper layers of the soil and removes comparatively smaller quantities of nutrients. Hence, the succeeding crop is not affected and is able to tap on the unused pool of nutrients in the surface layer of the soil.

Legumes benefitting the following crops are too well known. The residues and the root nodules release N during decomposition for the use of the following crop. It is also known that legumes absorb soil P more efficiently and part of this mobilized P in organic form is available to the succeeding crop. As a result, inclusion of legumes in any intensive cropping system has become imperative.

What is the best cropping sequence for the farm?

If a farm has been in continuous wheat for many seasons and yields are poor or declining, the introduction of alternative species (break crops) into the cropping sequence might boost yield considerably. Interactions between the wheat and break crops are complex because there are variable roles of nitrogen, water, disease and weather. This chapter examines these complexities. It explores how and why they might influence the performance of wheat in different sequences and how they might impact on the economics of the farming system. It then outlines trials to

test these interactions. These three-to four-year trials when complete should provide the basis for crop sequence management packages for the local region.

Which farms could benefit from these trials?

- Those farms that have cropped wheat continuously for many seasons;
- those farms with inexplicably declining yields;
- those with ‘whiteheads’ during grain filling;
- those farms with root diseases, nematode problems and generally poor growth;
- those consistently producing small and shrivelled grain not attributed to drought;
- those located near markets for alternative crops and managed by adventurous farmers who have the foresight to diversify.

Terminology: sequences versus rotations

This chapter deals with crop sequences not rotations. In rotations different crops are grown in a fixed order year after year but in sequences the order is not necessarily repeated. The term ‘sequence’ captures the flexibility that is a feature of aware farm management, continuously responding to changing market demands and changing on-farm constraints.Here the focus is on short- to medium-term crop sequence trials rather than the fully phased long-term rotations traditionally conducted on experimental stations.

Background to cropping sequencing: Generally, the yields of crops grown in continuous monoculture decline as a result of a build-up of soil or stubble-borne disease that is specific to that crop species.

This problem can be overcome in some cases by the use of varieties that are resistant to the disease, but for some soil-borne diseases there is no current genetic resistance. Chemical control is usually not economical so that the only control strategy for these diseases is to grow a non-host or break crop that is not infected by the disease, so that the levels of disease in the soil are reduced prior to

the next wheat crop. Crop rotation of this sort has been practised for thousands of years. As well as reducing the levels of cereal disease, legume break crops, such as peas, lupins and lentils, also fix nitrogen from the air. This provides a beneficial N contribution to following cereals that may persist for longer than one year.

Finding an appropriate sequence crop

The challenge is to find break crops that farmers are willing to include in their crop sequence. Farmer adoption will usually depend on the following:

- Can farmers readily grow the crop? Do they have the equipment, knowledge and skills?
- Is there an accessible market for the product?
- Is the crop as profitable as the wheat crop it replaces?
- Will there be other benefits to productivity of the whole farm in the longer-term?

Break crops must be adapted to the soils and climate of the area and be accompanied by a good agronomic management package if they are to be adopted successfully by farmers.

What influences a crop's response to preceding break crops?

Disease control

The most significant benefit of break crops to following cereals is the reduction in soil and stubble-borne disease. In Australia for example, the yield of wheat after canola or linseed crops has averaged 20 percent higher than for wheat, after wheat largely because of disease reduction.

Impact of a break crop on the incidence of 'whiteheads' in a following wheat crop



Nitrogen benefits

In addition to providing a disease break, grain legume break crops such as peas, lupins, lentils and chickpeas may provide residual N to following wheat crops. In some regions these benefits have boosted yields of following wheat crops by up to 50 percent, similar to the effects of applying adequate N fertilizer out of a bag. However, the magnitude of the N benefit depends on effective N fixation by the legume, the amount of biomass produced and how much N is removed in legume seeds or stubble.

Some non-legume break crops, such as linseed, may also leave any deep residual soil N untapped because they have a shallow root system and use soil N inefficiently. Since residual N influences both yield and protein content of following cereals, it is important to consider the N balance of the whole system when attempting to understand crop sequence effects (see chapter on *Optimizing nitrogen use on the farm*). Benefits in N nutrition may also arise from non-legume break crops simply because they effect a healthier wheat root system, enabling the wheat crop to use soil N and applied N fertilizer more efficiently and thereby yield better.

Water use by preceding crops

In drier environments or in dry seasons the amount of water removed from soils by preceding crops can have a major effect on the growth of following crops.

In annual cropping cycles the inclusion of deep-rooted rotation crops that remove significantly more water than wheat may cause yield loss in some seasons.

Impacts on nutrients other than N

Be aware of the nutrient requirements and likely removal by prospective break crops. *Brassica* crops such as canola, remove large amounts of elements from the soil, particularly S and Zn. These crops may induce deficiency in following crops if adequate amounts are not supplied.

Arbuscular mycorrhizal fungi (AMF) colonize the roots of most crop plants and symbiotically help acquire immobile elements such as P and Zn. In general, tap-rooted crop species are dependent on AMF for P and Zn nutrition in low P soils while fibrous rooted cereals are less dependent. Canola and some lupins are among the few species that are not hosts of AMF so that AMF inoculum levels in soil are usually low following these crops. If following crops in the sequence are AMF dependent or if your soils are low in P or Zn, you may need to apply the required nutrients.

Stubble allelopathy and herbicide history

Other interactions between crops in the sequence can arise from chemical interference caused by leached compounds from retained residues (allelopathy) or from residues of herbicides applied to the previous crop.

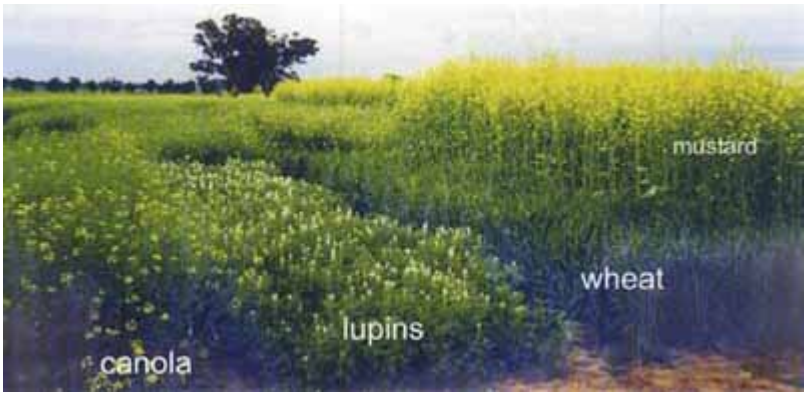
Seasonal interactions

All the above-mentioned factors influenced by crop sequence also interact strongly with seasonal conditions. The complexity of the many possible interactions can lead to outcomes that may be difficult to understand or explain. For example, dry periods between seasons increase carry over of soil-borne diseases (dry conditions maintain inoculum levels). Root infection is increased by wet, cool

conditions during early growth while dry conditions during grain fill, allow expression of the diseases as “whiteheads”. These heads have small pinched grains or no grains. Alternatively, wet periods prior to sowing may negate any differences in disease inoculum. This may be by reducing inoculum directly. However, it may also be via the indirect route of improving the following crop by refilling the profile with water and mineralizing large amounts of soil N, so making that crop less susceptible to disease.

Such seasonal ‘weather’ interactions are largely responsible for variations in response to preceding crops. So although published studies have shown an average response of wheat yield to non-legume break crops of a 20 percent increase, this spans a *104 percent increase to a 12 percent decrease*. As there are so many complex variations and interactions, the only way to get a reliable picture across your region is to actually conduct crop sequence trials at several sites over several seasons. The human resource needed will clearly be considerable, ideally being interested, collaborating farmers working on their farms. Each farmer will benefit by defining together with you the appropriate sequence for their location while you, as coordinator, will learn the driving forces for the region. The overview information should give you confidence to design sequence packages for any site within the region that will have a high chance of working.

Some subtreatments in one replicate of year 1 in the case study



Quantifying the advantages of sequences using a gross margins analysis

A good place to start assessing a proposed crop sequence is to do a rough cost and benefits analysis (gross margins). Decide on the crops you might use in the sequence then sum the likely income from each crop in the several-year sequence. Next, subtract the variable costs from that sum (seed, fertilizer, labour, fuel and machinery wear). Compare the result with a sum calculated for continuous wheat crops. How much would the yield of the wheat crop in the sequence have to improve to make an overall improved profit compared with continuous wheat? In some sequences, reduced direct income from the break crop might be offset by an increased income from a higher yielding wheat crop and/or by a reduced requirement for N fertilizer in the following year.

Other costs and benefits, which may be more difficult to quantify, are those associated with risk.

On the negative side, the risk of crop failure may be increased if the break crops are not well adapted to the region. On the positive side, having product diversification in years of cereal failure or low cereal prices may lessen risk. The costs and benefits of spreading sowing and harvest workloads of the different crops and on longer-term improvements in soil fertility are also difficult to assess.

Chapter 6. Indices for evaluation for cropping system

The most important index of biological advantage is the relative yield total (RYT) or land equivalent ratio.

1. The mixture yield of a component crop expressed as a portion of its yields as a sole crop from the same replacement series is the relative yields of the crop and sum of relative yields of component crop is called Relative yield total (RYT).

2. The total land area required under sole cropping to give the same yields obtained in the intercropping is called Land equivalent ratio (LER). Both the expressions (RYT and LER) are similar.

1. Land Equivalent Ratio (LER):

This is the most frequently used efficiently indicator. LER can be defined as the relative and sole crop that would be required to produce the equivalent yield achieved by intercropping.

$$LER = \frac{Yab}{Yaa} + \frac{Yba}{Ybb}$$

Where, Yab = yield of crop a in intercropping, Yba = yield of crop b in intercropping, Yaa = yield of crop a in pure stand and Ybb = yield of crop b in pure stand.

LER of more than 1 indicates yield advantage, equal to 1 indicates no grain or no gain or no loss and less than 1 indicates yield loss. It can be used both for replacement and additives series of intercropping.

LER is the summation of rations of yields of intercrop to the yield of sole crop.

Example: Let the yields of groundnut and red gram grown, as pure crops are 1,200 and 1,000 kg/ha, respectively. Let to yields of these cops when grown, as intercrop be 1,000 and 600 kg/ha, respectively. The land equivalent ratio of groundnut + red gram intercropping system is

$$\text{LER of groundnut} = \frac{\text{Yield of intercrop}}{\text{Yield of sole crop}} = \frac{1000}{1200}$$

$$\text{LER of red gram} = \frac{600}{1000}$$

$$\text{LER of system} = \frac{1000}{1,200} + \frac{600}{1000} = 1.43$$

LER of 1.43 indicates that a 43 percent yields advantage is obtained when grown as intercrop compared to growing as sole crops. In other words the sole crops have to be grown in 1.43 ha to get the same yields level that is obtained from 1.00ha off intercropping.

2. Relative Yields Total (RYT):

The mixture yields of a component crop expressed as a portion of its yields as a sole crop from the same replacement series is the relative yield of crop and sum of the relative yields of component crop is called Relative Yields total (RYT). Or When LER is compared at uniform overall plant density of sole and intercrops then it is known as RYT. In RYT yields advantages is to measure not only unit area, but also on unit population. This is mainly used for replacement series.

$$LER = \frac{Yab}{Yaa} + \frac{Yba}{Ybb}$$

Where Yaa = yields of component a as sole crop Ybb = yields of component b is a sole crop

Yab = yields of component a as intercrop in b Yba = Yield of component b as intercrop in a.

Example: In pasture mixture, stylo and anjan grown in 1:1 ratio with 50% sole crop population of both crops. In intercropping, mixture yield (50%) for stylo and anjan is 6 and 4 tons/ha of fodder, respectively. The yield of these crops in sole stand with 100% PP is 10 and 8 ha of green fodder, respectively.

RYT = 12+8/10+8 = 20/18 = 1.11.

So, RYT 1.11 indicates 11% extra fodder yields obtained by mixture.

Two crop yields in the intercrop mixture are MA and MB and the yield of crops grown of crop are SA and SB then the combined index is

$$L = MA + MB = \frac{LA}{MB} + \frac{LB}{SB}$$

The intercrop embodied in LER is that L represents the land required for sole crops to duce yields achieved in the intercropping mixture. A value of L greater than (1) indicates overall biological advantages of intercropping. The two component of total index LA and represent the efficiency of yields production of each crop when grow in mixture, relative to crop performance.

Relative Crowding Coefficient: The other index used is relative crowding coefficient which can define in terms of LER component as

$$\frac{LA}{1 - LA} \times \frac{LB}{1 - LB}$$

The two main indices of dominance are the aggressivity and competition index. aggressivity gives a simple measure of how much the relative yield increase in species. A is after than that species B. It is an index of dominance.

$$\text{Aggressivity} = \frac{\text{Mixture yields of A}}{\text{Expected yields of A}} - \frac{\text{Mixture yields of B}}{\text{Expected yields of B}}$$

$$= \frac{MA}{SA \times ZA} - \frac{MB}{SB \times ZB}$$

Sown proportion of species A and B are represented as ZA and ZB, respectively.

An aggressivity value zero indicates that the component species are equally competitive. A positive sign indicates the dominant species and a negative sign the dominated.

The basic process in the competition index is the calculation of two equivalence factors, one each component species. It is the product of two equivalence factors, one for each component species.

Effective Land Equivalent Ratio (ELER): Any required ratio could be achieved by growing the intercrop of the land area and one of sole crops the remainder.

$$P = \frac{(La + E)}{(LER + E)}$$

LER is a measure of the net advantage from the combined intercrop determined by adding the crop and sole crop relative yields.

ELER = Lb /((1- La) + (LER – 1) P)

Staple Land Equivalent Ratio: In situation where the primary objectives is to produce yields of one component (staple crop) usually a cereal and some yields of the legume.

3. Land Equivalent Coefficient (LEC):

It was proposed by Adetilaye and Ezedinma (1986) the product of LER of intercrop components.

$$= \frac{MA \times MB}{SA \times SB}$$

It can therefore, be regarded as a measure of association or interaction when crop with the strength of relationship for a two crop mixture the minimum expected product by yield advantage is obtained and LEC greater than 0.25 (25%). This was obtained from the 50:50 yields where interspecific competition equaled intraspecific competition.

LEC = LA x LB = 0.5 x 0.5 = 0.25.

The LEC is developed to assess the interaction and potential of crop mixture. It is derived from the understanding that the in intercrop yields, as compare optimum sole crop yield is due competition assuming that the inter competition is absent.

4. Area Harvest Equivalency Ratio (AHER):

This was proposed by Bal Subramanian and Sekayange (1990). It indicates resources efficiency. The concept of AHER combines the area time factors in a practical sent quantifying intercrop yield advantage particularly in multi season.

Land Use Efficiency or Assessment of Land Use:

The main objective is to use available resource effectively. Multiple cropping which include both inter and sequential cropping has the main objective of intensification of cropping with the available resources in a given environment. Several indices have been proposed to compare the efficiencies of different multiple cropping system in turns of land use, and these have been reviewed by Menegay et al. 1978.

1. Multiple Cropping Index or Multiple Cropping Intensity (MCI):

It was proposed by Dalarymple (1971). It is the ratio of total area cropped in a year to the land area available for cultivation and expressed in percentage.

$$MCI = ai/A \times 100$$

Where i = 1, 2, 3, n, n= total number of crops, ai = area occupied by crop and A= total land area available for cultivation. Or MCI is the sum of area planted to different crops and harvested in a single year divided by total cultivable area and expressed as percentage. Or MCI means the sum of areas under various crops raised in a single years divided by net area available for that cropping pattern multiplied by 100. It is similar to cropping intensity.

$$MCI = \frac{\text{Total number of crops + with their respective area}}{\text{Net cultivable area}} \times 100$$

2. Cultivated Land /Utilization Index (CLUI):

Cultivated land utilization Index (Chuang, 1973) is calculated by summing the products of land area to each crop, multiplied by the actual duration of that crop divided by the total cultivated land times 365 days.

$$CLUI = \frac{aidi}{Ax365} \times 100$$

Where, i = 1, 2, 3,---- n, n = total number of crops. ai = area occupied by the ith crop, di = days that the ith crop occupied and A = total cultivated land area available for 365 days.

CLUI can be expressed as a fraction or percentage. This gives an idea about how the land area has been put into use. If the index is 1 (100%), it shows that the land has been left fallow and more than 1, tells the specification of intercropping and relay cropping. limitation of CLUI is its inability to consider the land temporarily available to the farmer for cultivation.

3. Crop Intensity Index (CII):

Crop intensity index assesses farmers actual land use in area and time relationship for each crop or group of crops compared to the total available land area and time, including land that is temporarily available for cultivation. It is calculated by summing the product of area and duration of each crop divided by the product of farmers total available cultivated land area and time periods plus the sum of the temporarily available land area with the time of these land areas actually put into use (Menegay et al. 1978). The basic concept of CLUI and CII are similar. However, the latter offers more flexibility when combined with appropriate sampling procedure for determining and evaluating vegetable production and cropping pattern data.

$$CII = \frac{ai.ti}{A_OT + M\,(AiTj)}$$

Where, i = 1, 2, 3……NC, NC = total number of crops grown by a farmer during the time period. T, ai = area occupied by i^{th} crop (months that the crop i occupied an area ai) T = time period under study (usually one year), A_O= Total cultivated land area available with the farmer for use during the entire time period, T, M= total number of fields temporarily available to the farmer for cropping during time period. T, j=1, 2, 3…….. M, Aj=land area of jth field and Tj= time

period Aj is available. When, CII = 1 means that area or land resources have been fully utilized and less than 1 indicates under of resources.

CII indicates the number of times a field is grown with crops in a year. It is calculated by dividing gross cropped area with net area available in the farm, region or country multiplied by 100.

$$CII = \frac{\text{Gross cropped area}}{\text{Net area}} \times 100$$

When long duration crop grown crop remain longer time in field this is the drawback of CII. So, time is not considered, thus , when long duration crops like sugarcane and cotton are grown, the cropping intensity will be low, though crop remain longer period of time in the field.

4. Specific Crop Intensity index:

Also proposed by Menegay *et al*. 1978 SCII is a derivative of CII and determines the amount of area – time denoted to each crop or group of crops compared to total available to the farmers.

$$SCII = \frac{ak.tk}{A_O T + ai\ ti}$$

Where Nk= total number of crops within a specific designation such as vegetable crops or field crops grown by the farmer during the time period T. ak = area occupied by the kth crop. tk= duration of kth crop.

Using this formula vegetable intensity index, rice intensity index, field crops intensity index etc. can be calculated.

5. Diversity Index (DI):

It was suggested by Strout (1975). It measures the multiplicity of crops or farm products which are planted in a single year by computing reciprocal sum of squares of the share of gross revenue received from each individual farm enterprises in a single year.

$$DI = \frac{1}{Yi}$$

Where,

N = Total number of enterprises (crops or farm products) and yi = gross revenue of ith enterprises produced within a year.

6. Harvest Diversity Index (HDI):

It is computed using the same equation as the DI expects that the value of each farm enterprises is replaced by the value of each harvest.

$$HDI = \frac{1}{Yi}$$

Where,

Yi = gross value of ith crop planted and harvested within a year.

7. Simultaneous Cropping Index (SCI):

It is computed by multiplying the HDi with 10,000 and dividing the product by MCI. (Strout, 1975).

$$SCI = \frac{HDIx10,000}{MCI}$$

8. Relative Cropping Intensity Index (RCII):

It is the modification of CII and determines the amount of area and time allotted to 1 crop or groups of crop related to area- time actually used in the production of all crops. RCII numerator equal SCII denominator and RCII denominator equal CII numerator.

$$RCII = \frac{ak.tk}{ai.ti}$$

These indices can be used for classifying (i.e a farmer with relative vegetable intensity index 50% would be considered a vegetable grower) measuring shifts of various crops among farm of different sizes and determining whether the consistent types of cropping pattern occur within various farm size strata. These indices also held to know how intensity cultivated land area has been utilized. But none of these indices takes productivity into account and cannot be used for comparing different cropping systems and evaluating their efficiency in utilization of the resources other than the land.

9. Crop Equivalent Yield (CEY):

Many types of crops/ cultivars are included in a multiple cropping sequences. It is very difficult to compare the economic produce of one crop to another. To cite an example, yield of rice cannot be compared with the yield of grain cereals or pulse crops and so on. In such situations, comparisons can be made based on economic returns (gross or net returns). The yield of protein and carbohydrate equivalent can also be calculated for valid comparison, Effort have also been made to convert the yields of different crops into equivalent yield of any one crop such as wheat equivalent yield.

Crop equivalent yields (CEY): The yields of different intercrops are converted into equivalent yield of any one crop based on price of the produce.

Crop Performance Ratio and Its Types

It is defined as the productivity of an intercrop per unit area of ground compared was expected from sole crops sown in the same proportions (Azam Ali et al 1990). For each same then productivity in the intercrop can be expressed as a partial CPR.

Economic Viability:

The indicators like CEY, LER, RYT etc. give biological suitability of cropping system to an area. At the same time, cropping system should be economically viable and profitable following economic indicates can be used to evaluate profitability of cropping system.

1. Gross Returns:

The total monetary returns of the economic produce such as grain, tuber, bulb, fruit, etc. and byproducts viz. straw, fodder, fuel etc. obtained from the crops included in the system are calculated based on the local market prices. The total return is expressed in terms of unit area, usually one hectare.

The main draw back in this calculation is that market price of the produce is higher than that actually obtained by the farmer. Generally gross return calculated is somewhat inflated compared to the actual receipt obtained by the farmer.

2. Net returns or net profit:

This is worked out by subtracting the total cost of cultivation from the returns. This value gives the actual profit obtained by the farmer. In this type of calculation only the variable costs are considered. Fixed costs such as rent for the land, land revenue, interest on capital etc. are not included. For a realistic estimate, however, fixed costs should also be included.

3. Return Per Rupee Invested:

This is also called benefit-cost-ratio or input-m output ratio.

$$\text{Return per rupee invested} = \frac{\text{Gross return}}{\text{Cost of cultivation}}$$

This index provides an estimate of the benefit derived and expenditure incurred by the farmer in adopting a particular cropping system. Anything above the value of 2.0 (meaning that the farmer can get Rs.2 as return for every rupee invested) can be considered worthwhile.

4. Per Day Return:

This is called as income per day and can be obtained by dividing the net return by number of cropping period (days).

$$\text{Per day return} = \frac{\text{Net returns}}{\text{Cropping period (days)}}$$

This gives the efficiency of the cropping system in terms of monetary value. If the system is stretched over one year, the denominator can be replaced by 365 days and per day for the whole year can be calculated.

No single index is capable of giving good comparison of different cropping system and so a number of indices are used together to assess the economic viability of the system.

Chapter 7. Agronomic requirements in management of Cropping System

Agronomic considerations for different cropping system are different due to inclusion of more than one crop as in intercropping or sequence cropping system. Thus, principal involved in management of intercropping and sequence systems are different.

Intercrop is practiced with two objectives:

1) To get additional yields through and intercrop as bonus.
2) To avoid risk.

Management of Intercropping Systems:

In intercropping system crops are grown simultaneously. Management practices aim to provide favorable environment to all the component, exploit favorable interaction among the component crops and minimize competition among the components.

1. Seedbed Preparation: The objective of land preparation is to establish an ideal zone for the seedling that minimizes the stress. Potential stress condition include inadequate or excess moisture, unfavorable temperature for a given species, soil crushing weeds, residue of preceding crop and insect or pathogen attack. Importance of seedbed is the same in both conventional (monoculture) and in multiple cropping. Seedbed preparation depends on the crop. Deep rooted crops responds to deep ploughing while for most of cereal shallow tillage is sufficient. The crops with small seed required fine seedbed, cotton, and maize, planted on ridges, certain crops on flat seedbed. Since more than one crop in intercropping, the seedbed preparation is generally prepared as per the needs of base crop. Sugarcane planted in furrow and intercrop sown on ridges, certain crops on flat seedbed. Since more than one crop in intercropping, the seedbed preparation is generally prepared as per the needs of base crop. Sugarcane planted in furrow and intercrop sown on ridges. In Groundnut + red gram intercropping system, flat seedbed is prepared for sowing crops. However, ICRISAT is recommending broad bed and furrow for black soils. In rice + maize intercropping system, ridges and trenches are formed. Maize is planted on ridges and rice in trenches.

2. Varieties: The varieties of component crop in intercropping system should be less competing with the base crop and peak nutrient demand period should different from the base crop. Maturity period between two components is minimum of 30 days. Hybrids varieties of sorghum like CSH -6, CSH -9 are suitable for

intercropping with long duration variety of red gram like C11 and LRG 30 because of wider gap between maturity periods. The varieties selected for intercrop should be have thin leaves, tolerant to shading and less branching. If the base crop is shorter than intercrop, the intercrop should be compact with erect branching and its early growth should be slow. The characteristics of the base crop should be as in sole crop.

3. Sowing: Practices of sowing are slightly altered to accommodate inter- crop in such a way that it cause less competition to the base crop. Widening inter row spacing of cereal component to accommodate more rows of component legume crop improves legume yield and efficiency of the intercrop system. Sowing of base crop is done either as paired row, paired – winder row or skip row of base crop are brought close by reducing inter row spacing. The spacing between two pairs of rows is increased to accommodate the inter crop. Such row arrangement of component crops within the rows improves the amount of light transmitted to the lower component crop, which can enhance legume yield in cereal legume intercropping system.

For example – the normal row spacing of rainfed is 30 cm, the row spacing is reduced to 20 cm between paired rows and 50 cm spacing in two in two pairs. The spacing in paired row planting designed as 20/50 cm indicated that the spacing between two rows in pairs is 20 cm and among the pairs 50 cm.

Similarly, pearl millet is planted with row spacing 30/60 cm in paired row planting. These changes in crop geometry do not alter the yield of base crop, but intercrops are benefited to some extent. Or when alternating pairs of sorghum rows 90 cm with two rows of an associated legume, Singh (1972) found that LER was greater compared at 60 cm between rows with two rows of the legume in between Planting in fixed ratio of intercrop is most common. The intercropping system of groundnut + red gram is either in 5:1 or 7:1 ratio and sorghum + red gram in 2:1 ratio. In these cases the normal three tined or four seed drill can be used without any modification. For higher yields, base crop population is maintained at its sole

crop population and intercrop population is kept at 80 per cent of its sole crop population.

Relative sowing time of component crop is important management variable manipulated in cereal- legume system but has not been extensively studied. Sowing may be staggered to increase the temporal difference, which might result in higher yield advantage.

4. Fertilizer Application: The nutrient uptake is generally more in intercropping system compared to pure crops. When the legume are associated with cereal crop in intercropping system, legume supplement a portion of nitrogen required of cereal crop; the amount may be of 20 kg/ha by legumes. Application of higher dose of nitrogen to the cereal + legume intercropping system not only reduce the nitrogen fixation capacity of legumes, but also growth of legume is suppressed by aggressive cereals owing to fast growth of cereals with increased availability of nitrogen, cereal + legume intercropping, therefore is mainly advantageous under low fertilizer application.

Considering all the factors, it is suggested that the nitrogen dose recommended for base crop as pure crop is sufficient for intercropping system with cereal + legume or legume + legume. With regards to phosphorus and potassium, one eighth to one fourth of the recommended dose of intercrop is also added in addition to recommended dose of base crops to meet the extra demand. Basal dose of nitrogen is applied to rows of both components in cereal + legume inter crop. Top dressing of nitrogen is done only cereal rows. P & K are applied as basal dose to both crops.

5. Water Requirement: The technique of water management is the same for sole cropping and intercropping or sequential cropping. However, the presence of an addition crop may have an important effect on evapo- transpiration. With proper water management, it is possible to grow two crops where normally only one crop is raised under rain fed condition. Intercropping system is generally recommended

for rain fed crop get the stable yields. The total water requirement of intercrop does not increase much compared to sole cropping. At ICRISAT, the water requirement of sole sorghum and intercropping with red gram was almost similar (584 and 585 mm, respectively). However in a more competitive crop like onion as intercropped in groundnut increase the total water requirement by about 50mm. the total water used in intercropping system is almost same as in sole crops, but yields are increased. Thus water use efficiency of intercropping is higher than sole crops.

Scheduling of water: if one of the crop is irrigated based on its requirement, the other crop may suffer due to excess water stress, sometimes leading to total failure of crop. In cotton blackgram intercropping system, cotton is irrigated once in 15-20 days. The intercrop black gram is often affected by excess water and gives poor yield. In such situations, skip furrow method of irrigation is advocated. Scheduling irrigation at IW/CPE ratio of 0.60 to 0.80 or irrigation at one bar soil moisture tension is suitable for most of the systems.

6. Weed Management: Generally it is believed that intensive cropping reduces weed problems. Weed infestation depends on the crop, plant density and cultural operation done. Weed problems is less in intercropping system compared to the sole crops. This is due to more complete crop cover and high density available in intercrop cause severe competition with weeds and reduce weed growth. The weed suppressing ability of intercrop is dependent upon the component crops selected, genotype used, plant density adopted, proportion of component crops, their spatial arrangement and fertility moisture status of the soil.

Experiment carried out at ICRISAT, Hyderabad, indicated that there was 50- 75 % reduction in weed infestation by intercropping. Pigeon pea + sorghum intercropping system, which is extensively practiced in Karnataka, M.S and A.P is known to reduce weed intensity. The higher plant population used and also complete covering of the soil earlier in intercropping system reduces weed infestation. In late maturing crops that are planted wide rows, presence of early maturing crops helps to cover the maturing crops that are planted wide rows,

presence of early maturing crops helps to cover the vacant inters row space and keeps weed under check.

Quick growing non- competitive, compact legumes like green gram and black gram act as another crop due to their good canopy coverage. In certain situations, intercrops are used as biological agents to control weeds, black gram, green gram, cow pea in sorghum and cowpea in banana reduce weed population and one hand weeding can be avoided by this method, however, in some intercropping system like maize + groundnut, rice+ tapioca, maize + tapioca, weed problem is similar to their sole crops. The growth habit of genotype used in intercropping has a great influence on weed growth. Weed infestation in intercropping is influenced by early growth and competitive additives of the component crops. If one or both the component crops are vigorous and cover the land area rapid ally, weed infestation is greatly reduced. Early crop canopy to cover the soil is more important than rapid increase in plant height.

It is well known that, different species of weeds are associated with different crops, but weeds present in sole crops are different than those present in intercropping system. At Hyderabad, in pearl millet as sole crop mixed weed flora was observed as *Celosia, Digitaria* and *Cupreous* in sole crop of groundnut. In pearl millet + groundnut intercropping system type of weeds changes with proportion of component crops. As more rows of groundnut are introduced in place of pearl millet of rows, there is a striking increase in both numbers and biomass of the tall and competitive Celosia, especially in groundnut rows.

Weed problem is less; weed control is necessary in intercropping system. But labour required for weeding is less; second weeding is not necessary because of crop coverage and limited weed growth. Normally two hand weeding are required, but it may restrict to one hand weeding under intercropping in sorghum + red gram or sorghum + cowpea, just one weeding is sufficient to get high yield as in weed-free check.

The critical period of weed free condition may be extended a little longer in intercropping than in sole cropping. This is because the critical growth stages of the component crops very temporally in intercropping for example, critical weed free period has to be extended to first 7 weeks in sorghum + red gram intercropping while sole sorghum crop requires only requires only 2- 4 weeks weed free period.

Chemical weed control is difficult in intercropping system because the herbicide may be selective to one crop but non- selective to another. Atrazine control weeds in sole sorghum, but it is not suitable for sorghum + red gram intercropping system, as it is toxic to red gram. Herbicides suitable for intercropping systems as-

* Maize + green gram & Maize + cowpea- Butachlor (pre- emergence) (Machete)
* Sorghum + pulse – fluchloriam (PPI) (Basalin) –or Alacholor (pre- emergence) (loss)
* Sorghum + red gram – prometryne (pre- emergence)
* Sugarcane + groundnut – nitro fen (pre-emergence) (to KE -25).

7. Pest and disease in intercropping System: Pest and diseases are believed to be less in intercropping system due to crop diversity than sole crops. Some plant combination may enhance soil fungicide and antibiotics through indirect effects on soil organic matter content. The spread of the diseases is altered by the presence of different crops. Little leaf of brinjal is less when brinjal is sheltered by maize or sorghum, as the insect- carrying virus first attacks maize or sorghum; virus infestation is less on brinjal. Non – host plant in mixtures may emit chemicals or odor that affects the pests, thereby protecting host plants.

The concept of crop diversification for the management of nematode population has been applied mainly in the form of decoy and trap crops.

Decoy crops are non-host crops, which are planted to make nematode waste their infection potential. This is affected by activating larva of nematode in the absence of hosts by the decoy crops.

Crop	Nematode	Decoy crops
Brinjal	*Meloidogyne incognita*	*Sesamum orientale*
Tomato	*Meloidogyne pratylenchus alleni*	Caster, Groundnut
Soybean	*Pratylenchus sp.*	*Crotalarias spectabills*

Trap crops are host crops sown to attract nematode but destined but destined to be harvest or destroyed before the nematode manage to hatch. This is advocated for cryst nematode. The technique involves sowing in pineapple plantations; tomatoes are planted and ploughed in to reduce root knot nematodes. There is also evidence that, some plants adversely affect nematode population through toxic action. Marigold reduces the population of pratylenchus species worms.

Management of Sequential Cropping System

Unlike intercropping crops are grown one after another in sequential cropping and hence, management practices are different.

1. Seedbed Preparation:

Suitable can be prepared as per crops. Puddling for rice, ridges and furrow for vegetable, maize and cotton and flat seedbed for several other crops. However, two problems are encountered in seedbed preparation in sequential cropping system.

1) The time available for seedbed preparation is less in high intensity cropping system. Frequent rain interference the land preparation.

2) Due to prevent crop field may be in condition. For example field preparation after rice is difficult, it is mainly because soil structure is destroyed during puddling.

The turn – around time, the time between harvesting of crop to sowing of next crop in more if rice is the preceding crop. To avoid this problem minimum

tillage or zero tillage is adopted. It is the common practices to sow pulse crop just before or immediately after harvesting. Rice crop in rice- wheat system the stubbles are killed by spraying paraquat and wheat is sown in plough furrows between stubble of rice.

In irrigated agriculture the time available for tillage between two successive crops is minimal leading to minimum tillage. Minimum tillage is applicable for soils with,

1. Course texture surface soil
2. Good drainage
3. High biological activity of soil fauna.
4. An adequate quantity of soil residue mulch.
5. Favourable initial soil moisture and friable soil consistency over a wide range of soil moisture.

It is not possible to practice zero or minimum tillage in all sequence cropping systems. If sunflower is the preceding crop, ploughing is essential to oxidize the allelo chemical of sunflower. The stubbles of pearl millet and sorghum, which contain high C:N ratio immobilizes nitrogen. It is therefore, necessary to remove them. Stubble also interfere the operations.

In rice-rice-green gram system summer ploughing is done and later when water is available in the rainy season puddling is done and first crop is sown. For the second rice crop minimum tillage as incorporation of crop residues and rice is planted. Green gram is sown as a relay crop in the second rice crop.

In Cotton-Sorghum-finger millet cropping system in garden lands, thorough field preparation is done and field is laid out into check basins to transplant finger millet. In next season, cotton is planted among the stubbles of finger millet without any kind of preparation. Weeds are controlled by intercultivation operation in Nigeria, no till planting of sorghum into residues of the

previous crop maintained the seed at 10°C lower temperature in the seedling zone reaches 41° C.

2. Varieties:

Short duration of crops is selected to fit well in the multiple cropping systems. Photo- insensitive varieties are essential for successful sequence cropping system.

3. Sowing:

Sowing is not a problem as provided sufficient time for seedbed preparation. If seedbed not prepared well, difficult the establishment of crop for example- cotton establishment is difficult in black soil after rice. Due to hard pans in the shallow layer, root penetration is difficult. If field allowed to dry for land preparation, lost of one month, hence seedling are raised either on twisted paddy straw of leaf cups before harvesting of rice crop. After harvesting a cow bar hole is made up to 30 cm. it is partly filled with sand and soil mixture and cotton seedling are planted. The establishment of pulse crop is difficult after rice broadcasting of seeds rice crop or rice subjected results uneven germination and high seed rate is necessary. Crops planted in stubbles are subjected to competition from regenerated stubbles. It can be overcome by spraying preparation or digger on stubbles.

Delay in sowing is common problems in intensive cropping systems. To reduce yields loss due to transpiration of overage seedlings higher level of nitrogen is applied to induce tillering. In rice – wheat system, wheat yields are reduced considerably when the sowing of wheat is delayed beyond November. In such situation, transplanting of 40 to 50 days old seedlings of wheat is done. FYM is broadcasted over the field to maintain higher soil temperature during December.

4. Soil fertility management:

Soil fertility management become more complex in intensive cropping because of the additional factors such as residual effect of nutrients applied to the previous crops, possible effect of legume in the system, complementary and competitive interaction from the component crops and influence crop residues left in the soil. The modern or chemical agriculture, which includes higher cropping intensity involving improved varieties, heavier inputs of fertilizer and water, increasing yields and accelerated removal of plant nutrients have inputs of fertilizer and water, increasing yields and accelerated removal of plant nutrients have added newer dimensions to the fertility management. Fertilizer practices for sequence cropping: based on long term fertility experiments conducted in various parts of India, the following broad conclusions are:

System productivity increased with the application of P along with N, and further increased with use of N, P and K, Application of N at recommended dose is advocated to each of the intercropping system.

Phosphorus management in cropping system needs careful adjustment of P fertilizer dose taking into the account type of fertilizer, soil characteristics and their yield level, extent of P removal and growing environment. In cropping system – involving wheat, fertilizer P dose to Kharif crop can be reduced if preceding wheat has received P in adequate amounts.

Removal of K in proportion to N is very high in cropping systems particularly those involving cereal and fodder crops. It is important to apply K fertilizer are recommended dose to maintain soil fertility. In K rich soils of Coimbatore, K application at 50% recommended dose to each crops in the sequence rice-rice-soybean was optimum. Among the secondary nutrients, sulphur application is benefited particularly to sesamum- mustard, soybean- safflower and groundnut- mustard cropping sequence (Tandon, 1991). Among micronutrients, Zn deficiency is the most common as nearly 50% soils of intensive cultivated areas suffers from Zn deficiency. Rainy season crops like rice, maize and sorghum respond more to applied Zn than winter crops like wheat and chickpea. Under long

term experiments at Ludhiana (Punjab), significant decline in productivity of maize was recorded after 10 to 12 years' annual cycles, due to fall in Zn status. Use of organics prevents Zn deficiency in intensive cropping system under normal soils.

5. Cropping System and Integrated Nutrient Management (INM):

The concept of integrated nutrients management (INM) involves use of various inorganic, organic, biological sources of nutrient for improvement and maintenance of soil fertility leading to sustained crop production. Crop responses to organic and biological sources of nutrients for improvement and maintenance of soil fertility leading to sustained crop production. Crop responses to organic and biological sources of nutrients are not spectacular as to fertilizers, but the supplementary and complementary use of these resources is known to enhance the use efficiency of applied fertilizer besides improving soil physico- chemical properties and preventing emergence of micro – nutrient deficiencies.

The major components of the INM are fertilizer, organic manures, green manures, crop residues and bio fertilizers. In cereal- based cropping systems, about 25-50% fertilizer NPK dose of rainy season crops could be curtailed with the use of organics such as FYM, green manure and crop residues. In sugarcane based system, integrated use of press- mud, cane trash and bio fertilizers each with inorganic fertilizers and green leaf manuring showed 20-25% economy of fertilizers N applied to sugarcane by improving the use efficiency of N, P and other nutrient.

Effect of intensive cropping on soil properties: Physical properties- continuous ground cover due to intensive cropping minimizes soil erosion, run-off losses and crust formation. Relatively higher amount of crop residues due to intensive cropping improves soil structure.

Chemical properties- Organic residues in intensive cropping systems should be recycled to maintain optimum organic carbon in the soil for sustained production.

Chapter 8. Sustainable agriculture, role of farming systems in sustainable agriculture

In India progress in agriculture was through exploitation of recourses to serve the interest of rules in the power. In the post- independence era, we were faced with the difficult task of feeding an increasing population and were burned by recurring import of food grains. Attending self- sufficiency in food grain production was the challenges and a goal before agricultural scientists. The element of strategy to achieve self- sufficiency in food production within a short time were to expand the area under cropping, make large investments in development of water resources, develop input supplies and market infrastructure. Developments and

promotion of high yielding varieties responsive to high inputs was the cornerstone for green revolution.

We are self sufficient in food grain production. Now the question arises whether to follow the same strategies, which resulted in green revolutions or to redefine them. Consideration of our past achievement new challenges and the related development worldwide can answer this question.

1. Although we are now self sufficient in food grain production, the per capital availability has not increased significantly and a large fraction of population are still below poverty line due to lack of purchasing capacity.

2. At the present growth rate of 2 % per annum, we will be about 1.3 billion by 2025 and 1.6 billion by 2060, hence there is need to produce more from shrinking resources base.

3. Major gains in productivity and production in the past three decades have been from areas, which had no serious limitation to production (adequate irrigation productive soils, ideal climate etc.) productivity and production increases was moderate in eastern and central India and Deccan plateau.

4. Even in areas with no serious limitation for production, technology adopted for enhancing productivity has also simultaneous weaken the resource base resulting in a series of second – generation problems.

Modern crop production technology has considerably raised output but has created problems of land degradation, pesticide residue in farm produce, erosion, and atmosphere and water pollution. The natural resources base is degraded and diminished and the quality of the environment sustaining human life is adversely affected. Agricultural production has sustained man and great civilization. The history of the world reveals that great civilization flourished along irrigation sources and mismanagement of these resources saw the extinction of these civilizations. With expanding population and rapid depletion and degradation of the

natural resource base sustainable agriculture has assumed very great significance. The task of meeting the needs of the present generation without eroding the ecological assets of the future generation is receiving up priority by environmental planners.

The word sustainability is now widely used in development circles. But what does it really mean? According to a dictionary meaning sustainability refers to keeping an effort going continuously, the ability to last out and keep from falling. In the context of agriculture, sustainability basically refers to the capacity to remain productive while maintaining the resources base for example, the Technical Advisory Committee of the Consultative group on International Agricultural Research (TAC/CGIAR). Sustainable agriculture is the successful management of resources for agriculture to satisfy human changing needs, while maintaining of enhancing the quality of the environments and conserving natural resources.

Judging Agriculture to be Sustainable if it is:

1. Ecological sound: which means that the quality of natural resources is maintained and the vitality of the entire agro-ecosystem form humans, crop and animal to soil organism is enhanced.
2. Economically viable: which means that farmers can produce enough for self-sufficient and or income is measured not only in terms of direct farm produce (yield) but also in terms of function such as conservation resources and minimizes risks.
3. Socially just which means that resources and power are distributed in such a way that the basic needs of all the members of society are met and their rights to land use, adequate capital technical assistant and market opportunities are assured.
4. Humane: Which means that all forms of life (plant, animal, human) are respected.
5. Adaptable: Which means that rural communities are capable of adjusting to the constantly changing condition for farming populating growth, policies, market demand etc.

Definition: There are many definitions of sustainable agriculture. Simply stated “Sustainable agriculture is a form of agriculture aimed at meeting the needs of the present generation without enhancing the resource base of the future generation.

In order to feed the burgeoning population more food has to be produced and this to be done without enhancing the resources base. Expanding agriculture to ecologically fragile area means greater threat to environment. According to some workers sustainable agriculture is minimal dependence on synthetic fertilizer, pesticide and antibiotics. It is also considered as a system of cultivation with the use of manure, crop rotation and minimal tillage.

There is an urgent need to develop farming techniques, which are sustainable from environment and socioeconomic, points of view. In 1980s, We Jackson of the Land Institute in Salina using the term the sustainable agriculture to describe an alternative system of agriculture based upon resources conservation and quality of rural life. The sustainable agriculture movement started in 1980.

Sustainable agriculture emphasizes the conserving of his own resources. For a farm to be sustainable, it must produce adequate amounts of high quality foods, be environments safe and where appropriate, be profitable, sustainable farms minimizes their purchased inputs (fertilizers, energy and equipment) and rely, as much as possible on the renewable resources of the farms itself.

A group of Canadian scientists have defined sustainable agriculture as a philosophy and system of farming with it roots in a set of values that reflect a state of empowerment of a awareness of ecological and social realities and of one’s ability to take effective action which involves design and management procedures that work with natural processes to conserve all resources, promote agro-ecosystem resilience and self regulation and minimize waste and environmental impact, while maintaining or improving farm profitability.

Sustainable agriculture is a balanced is a balanced management system of renewable resources including soil. Wildlife forest crops, fish, livestock, plant genetic resources and ecosystems without degradation and to provide food, livelihood for current and future generation maintaining or improve productive and ecosystem service of these resources.

Sustainable agriculture system has to be economically viable both in the short and long term perspective. Natural resources not only provide food, fiber, fuel and fodder but also perform ecosystem service such as detoxification of noxious chemicals within soils, purification of nutrient and control weeds, pest and diseases through biological and cultural methods.

Sustainable agriculture is also a known as eco-farming or natural farming or perm culture. It is known as eco- farming or organic farming or natural farming. It is known as eco- farming as ecological balance is given important. It is also called organic farming, as organic matter in the main source for nutrient management. But some scientists consider that it is a mis-concept to think that sustainable agriculture is farming without chemical inputs. It is considered by some as integrated, low input and highly productive farming system.

Dr. M.S Swaminathan started in his write up in the Hindu Survey of Indian agriculture 1989 that ecological sustainability involves that addition of a time dimension of infinity in development programmes based on the utilization of natural resources. Sustainable development is the 20th century passes port to a future of promise for all nations. It is a passport that must be issued to all of humankind. The future of the most important solar energy harvesting enterprise.

Principles of Sustainable Agriculture:

1. No fight with nature, but co-operation with them.
2. Living soil- maintain the soil healthy.
3. Lessen the use of outside and distant resources.

4. Diversity and adjustment: The diversity of all plants and animals should be preserved and increased.
5. Durable livelihood: durable lemmatization about how much resource can be utilized by man in an area.
6. Self- Dependence: The new knowledge and technology should increase self-confidence, which help for effecting planning of resources.

Component of Sustainable System:

A wide chain of critical environment threats is posing challenges to the mankind all over the world. Some of these are degeneration of soil, water, forest and marine resources essential for increasing production of food, fodder\fiber, fuel, timber etc. Degradation of the natural resources is the main issue threading sustainable developments of agriculture. Causes of resource base degradation: following are the main reasons of derogation of natural resources. Major component of sustainable system (Role of IFS in sustainable agriculture):

They are as follows:

i) Soil and water conservation to prevent degradation of soil productivity and lengthening crop growing season for optimum crop production.
ii) Efficient use of limited irrigation water: without leading to the problems of oil, salinity, alkalinity and high ground water tables.
iii) Crop rotation to mitigate weeds, diseases and insect problems to increase soil productivity and minimizes soil erosion.
iv) Integrated nutrient management that reduces the need for chemical fertilizer improves the soil health and minimizes environmental pollution by conjunctive use of organics, in organics and bio fertilizer.
v) Integrated pest management that reduce the need for agro- chemical by crop rotation, weather monitoring, use of resistant cultivators, planning time and biological pest control.
vi) Management systems to control weeds by preventive measure, tillage, timely intercultivation and crop rotation to improve plant health.

All such ecologically sound practices require more information, trained labour and management skills per unit production than current input intensive agriculture.

Difference between Sustainable and Modern Agriculture:

Particulars	Sustainable agriculture	Modern agriculture
Plant Nutrient	Farm yard manures, compost, vermicompost, green manure, bio-fertilizer and crop rotation are used.	Chemical fertilizers are used.
Pest control	Cultural methods, crop rotation and biological methods are used.	Toxic chemical are used.
Inputs	High diversity, renewable and biodegradable inputs are used	High productivity and low diverse chemicals are used fragile ecology.
Ecology	Stable ecology	Fragile ecology
Use of resources	The rate of extraction from forests, fisheries, underground water source other renewable do not exceed the rate of regeneration.	The rate of extraction exceeds threat of regeneration. Falling of trees, deforestation, overgrazing and pollution of water - bodied takes.
Quality of food material	Food material are safe	Food material contain toxic residue.

Sustainability through Farming Systems

Two farming systems have been proposed for enduring sustainability. They are:

Low external input sustainable Agriculture or Low input sustainable Agriculture (LEISA/LISA)

It means minimal use of external production inputs. In view of the limited access of most farmers to artificial external inputs, the limited value of these inputs under LEIA conditions, the ecological and social threats of ‘green revolution’ technology and the dangers of production on non -renewable energy sources, the strong emphasis on High External Input Agriculture (HEIA) in agricultural development must be questioned. However, it is also open to question whether it will be possible to raise world food production sufficiently without the use of such

external inputs. Besides, natural as opposed to artificial inputs can also have detrimental environmental effects.

LEISA is an option which is feasible for a large number of farmers and which can complement other forms of agricultural production. As most farmers are not in a position to use artificial inputs or can use them only in small quantities, it is necessary to concentrate on technologies that make efficient use of local resources. Also, those farmers who now practice HEIA could reduce contamination and costs and increase the efficiency of the external inputs by applying some LEISA techniques. It is important that the agro-ecological knowledge of both scientists and farmers can be applied, so that internal and external inputs can be combined in such a way that the natural resources are conserved and enhanced. Productivity and security are increased and negative environmental effects are avoided.

A. LEISA refers to those forms of agriculture that

• Seek to optimize the use of locally available resources by combining the different components of the farm system, *i.e.,* plants, animals, soil, water, climate and people, so that they complement each other and have the greatest possible synergetic effects.

• Seek ways of using external inputs only to the extent that they are needed to provide elements that are deficient in the ecosystem and to enhance available biological, physical and human resources. In using external inputs, attention is given mainly to maximum recycling and minimum detrimental impact on the environment.

• LEISA does not aim at maximum production of short duration but rather at a stable and adequate production level over the long term. LEISA seeks to maintain and, where possible, enhance the natural resources and make maximum use of natural processes. Where part of the production is marketed, opportunities are sought to regain the nutrients brought to the market.

Numerous developing countries are now implementing so-called structural adjustment programs that involve policies such as devaluation of exchange rates, reduction of government spending and intervention, reduction of subsidies and removal of price controls. In this way, the demand for imports is to be curtailed and the purchase of local goods stimulated, so as to reduce the balance of payment and government deficits and to promote national economic growth. LEISA appears to fit within this context, as it is less demanding on imports and credits than the conventional approach to agricultural development. At farm, regional and national level, LEISA implies the need for closely monitoring and carefully managing flows of nutrients, water and energy in order to achieve a balance at a high level of production.

Management principles include harvesting water and nutrients from the watershed, recycling nutrients within the farm, managing nutrient flow from farm to consumers and back again, using aquifer water judiciously and using renewable sources of energy. As these flows are not confined by farm boundaries, LEISA requires management not only at farm level but also at district, regional, national and even international levels. At each level, technologies are sought to make the flow cycle as short as possible and to balance the flows. LEISA incorporates the best components of indigenous farmers' knowledge and practices; ecologically sound agriculture developed elsewhere, conventional science and new approaches in science (*e.g.,* systems approach, agro-ecology, biotechnology). Thus, conventional science has served mainly HEIA, but the contributions could make to LEIA should be explored to the full. LEISA practices must be developed within each ecological and socioeconomic system. The specific strategies and techniques will vary accordingly and will be innumerable. The experience thus far of developing LEISA systems cannot provide universal, ready-made answers for the problems of farmers in other areas, but can provide some indications of principles and promising possibilities.

The process of combining local farmers' knowledge and skills with those of external agents to develop site-specific and socio economically adapted farming techniques has been given the name 'participatory Technology Development' (PTD). Farmers work together with professionals from outside their community (*e.g.*, extension workers, researchers etc.) in identifying, generating, testing and applying new techniques. PTD seeks to strengthen the existing experimental capacity of farmers, and to encourage continuation of the innovation process under local control. The experience of combining indigenous and scientific knowledge through a process of PTD indicates strongly that it is indeed possible to transform LEIA to LEISA (Low External-Input and Sustainable Agriculture). This approach to agricultural development appears to be better adapted to the needs and opportunities of LEIA farmers and to fit better into their cultural context than the conventional approach.

B. Sustainable agro ecosystems

An alternative to the chemical dependence is to maximize the contributions of bio diversity to pest control and nutrient cycling and to attain optimal productivity with minimal inputs. This approach capitalizes the adaptive features of traditional systems and incorporates additional advantages of conventional and innovative technology. It is important to recognize a strong link between the availability of organic matter and both bio diversity and nutrient cycling. The practice in many developing countries of removing organic matter from the land for fuel and other purposes is a serious constraint to long-term sustainability. The most sustainable farming practices and components of the man managed bio diversity can be developed only by understanding the functions of the agro ecosystem and low social and economic conditions of the farmers and their climatic and environmental impact upon overall crop and animal productivity. No matter how well the agro ecosystem functions biologically, it is sustainable only if it is socially and economically sound.

Advantages

• Production costs are low,
• Overall risk of the farmer is considerably reduced,
• Pollution of water is avoided,
• Healthy food very little or no pesticide residue is ensured,
• Ensure both short and long term profitability.

Disadvantages

Continuation of LEISA will perpetuate a vicious circle of “low input-low yields” which the third world countries with even increasing population cannot afford. The solution for this is the optimal input farming which will meet the requirement of sustainability with the promise of low input/unit of output. It lays emphasis on law of diminishing returns.

Agronomic measures for management of scare and costly inputs

Agronomic management becomes difficult due to mismanagement of scare and costly inputs like- soil, rainfall, irrigation water, fertilizer, seeds etc. Soil provides physical encourages to plants and acts as store house for water and nutrients.

Thus, soil and water are the most critical basic resources so as to conserve efficiently as possible:

1. Adoption of soil and water conservation methods.
2. Management of irrigation water.
3. Good quality seed must be used.
4. Utilize suitable fertilizer.
5. Method and time of fertilizer application.

Chapter 9. Integrated Farming systems, factors governing choice and size of enterprises and resource allocation

Farming systems research (FSR): Originates from recognizing the inter-dependence and inter relationships of natural environment within the farming system. In FSR the farmers by participating in the research process help in the identification of the research problems as well as in testing the possible solution.

Goals:

The growing concern on suitable development has led the FSR to compasses on sound management of farm resources to enhance farm productivity and reduce the degradation of environment quality or to develop sustainable land use, which will optimize farm resource, minimum degradation with consideration to regenerative capacity, increase income and employment for farm families and promote quality of life.

Principal Involved: The FSR approach includes:

a) Viewing the farm as a whole,

b) Identifying the farming system, the interacting component and delineating boundaries,

c) Systematically investigation of the nature and extent of interdependence among the enterprises and identifying constraint,

d) Applying the modern technical know- how to the system so as to make it yield optimum results,

e) Studying the equity gender income, employment and resources use efficiency

f) Dealing with the issue at integration level through analysis and solution of problems towards sustainable farming system development.

In the past decade, farming system research has emerged as a popular and major theme in international agricultural research. FSR evolved in Post- Green Revolution era with the growing perception of the failure of main stream agricultural research and extension institution to generate and disseminate technologies widely adopted by small and resources poor farmers. Clearly technology, even when sound by scientific standards, is of limited value if is not adopted. The diagnosis of the problems was that agricultural researches and development planners, the generators and disseminators of new technology, had employed a fundamentally top- down approach to technology development, which is not valid one. In response to this situation, FSR argued that:

1. Development of relevant and viable technology for small farmers must be grounded in a full knowledge of existing of the farming system and

2. Technology should be evaluated not solely in terms of its technical performance, but in terms of its conformity to the goals, needs and socio- economic condition of small farm system as well.

Therefore, FSR concept was developed in 1970 in response to the observation that groups of small – scale farm families operation on harsh environment were not benefiting from conventional agricultural research and extension strategies. The tem FSR in its broadest sense is any research that views

the farm in a holistic manner and considers interactions (between component and of components with environment) in the system.

Farming system research is a research method designated to understand farmer's priorities, strategies and resources allocation decisions. It is most often used in conjunction with on farm highly location specific research with multi and inter disciplinary in nature and uses a whole farm approach for improved technologies to enhance and stabilize agriculture production.

The Characteristics of Farming System Research

1. It is holistic or system oriented,
2. It is problem solving: involvement of farmers in problem identification and solving process,
3. It is farmer participatory,
4. It envisages location specific technology solutions,
5. It is for specific client group – small/ marginal farmer,
6. It adopts bottom up approach,
7. It compasses extensive on farm activities, collaboration between farmer and scientist,
8. It is gender sensitive,
9. It ultimate objective is sustainability,
10. It focuses on actual adoption,
11. It recognizes interdependence among multiple clients.

Points to be considered while choosing the enterprises for Integrated Farming System (IFS):

1. Soil and climatic feature of an area/ locality.
2. Resource availability with the farmers.
3. Present level of utilization of resources.
4. Economics of proposed integrated farming system.
5. Farmers managerial skill.

6. Social customs prevailing in the locality.

Integration of Enterprises

In agriculture, crop husbandry is the main activity. The income obtained from cropping is hardily sufficient to sustain the farm family throughout year. Activities such as dairying, poultry, fishculture, sericulture, biogas production, edible mushroom cultivation, agro- forestry, agri- horticulture etc. assume critical importance in supplementing their farm income. Assumed regular cash flow is possible when cropping is combined with other enterprises. Judicious combination of enterprises keeping in view of the environment condition of a locality will pay grater dividends. At the same time it will promote effective recycling of residues/ wastes.

Integrated farming system seems to be the answer to the problems of increasing food production. For increasing income and for improving nutrition of the small scale farmers with limited resources without any adverse effect on environment and agro- eco-system. In a cropping system the amount of byproducts can be as high as or higher than marketable produce. This may go to waste if not utilized in an animal waste component and to reduce the cost of production of the economic produce of component two and finally to entrance the net income of the farm is whole.

Livestock is the best complementary enterprises with cropping, especially during the adverse years. Installation of a biogas plant in crop- livestock system will make use of the wastes, at the same time provides valuable manure and gas for cooking and lighting. In a wetland farm, there are greater avenues for fishery, duck farming and buffalo rearing; utilizing the rice straw and mushroom production can be started.

Under irrigated condition (garden lands) inclusion of sericulture, poultry and piggery along with arable crop production is an accepted practice. The poultry component in this system can make use crop the grains produced in the farm as feed. Pigs are the unique components that can be reared with the waste which are

unfit for human consumption. In Rainfed farming, sheep and goat rearing form an integrated part of the landscape, sericulture can be introduced in Rainfed farming provided the climatic condition permit it.

Agro- forestry (silvi- culture and silvi- horticulture) is the other activities, which can be included under dry land condition. In the integrated system, selection of enterprise should be on the cardinal principal that there should be minimum competition and maximum complementary effect among the enterprises.

Determinants of farming system or factors governing choice and size of enterprises and resource allocation in IFS

There are three major groups of factors, which in combination determine the type of farming system employed by framers in a given region. These factors are illustrated in Fig. 1 and discussed below.

Factor A represent the physical and biological elements which set limits to the type of agricultural produce to be produced in the given region. The physical elements include land, soil quality, topography, climate, water, location, distance etc. The biological elements include crops and livestock physiology, diseases etc., which determine the potential farm enterprises. These elements can be altered by limited intervention by the farmers and scientists. For instance scientists can evolve improved production technology and farmers can adopt it partially or in full package.

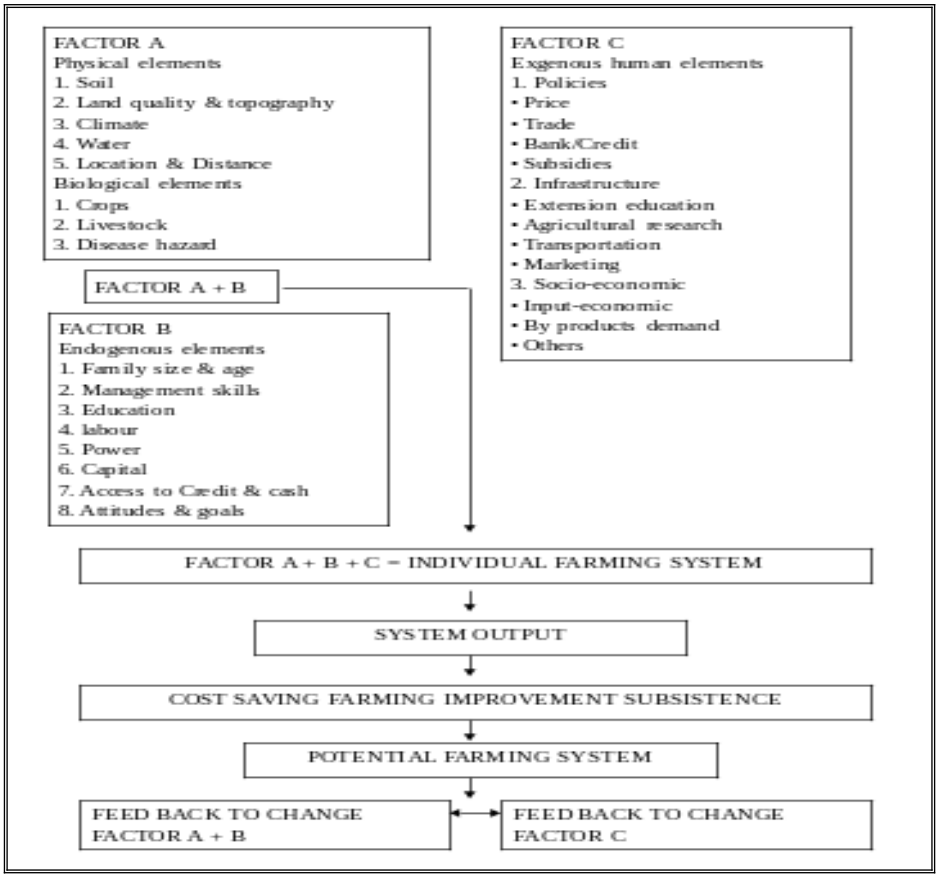


Figure: Determinants of farming system

Factor B represent endogenous human elements, which greatly influence the type of farming system adopted in a particular region. The system revolves around the farmer whose family and means of livelihood are intricately linked. The farm family has available resources under their control in terms of land, labour, capital and management. The quantity and quality of these resources are conditioned by the characteristics of the family (size, age etc), education and management skills, available labour, capital, power, attitudes and goals of the family. The farmer's goal and attitudes are initial factors that determine the nature of farming system specially where there is a range of alternative operations and enterprises to increase productivity consistent with existing technical elements. The farmer could combine available resources in a manner that will maximize the goals of the family.

Factor C represents the exogenous human variables, which govern the allocation of available resources by the farmers. Farm producers need incentives to change their farming methods and production patterns in desirable directions.

Chapter 10. Models of IFS for different agro-ecosystems

Benefits or Advantages of Integrated Farming System

1) Productivity: IFS provides an opportunity to increase economic yield per unit area per unit time by virtue of intensification of crop and allied enterprises.

2) Profitability: Use waste material of one component at the least cost. Thus reduction of cost of production and form the linkage of utilization of waste material, elimination of middleman interference in most input used. Working out net profit B/ C ratio is increased.

3) Potentiality or Sustainability: Organic supplementation through effective utilization of byproducts of linked component is done thus providing an opportunity to sustain the potentiality of production base for much longer periods.

4) Balanced Food: We link components of varied nature enabling to produce different sources of nutrition.

5) Environmental Safety: In IFFS waste materials are effectively recycled by linking appropriate components, thus minimize environment pollution.

6) Recycling: Effective recycling of waste material in IFFS.

7) Income Rounds the year: Due to interaction of enterprises with crops, eggs, milk, mushroom, honey, cocoons silkworm provides flow of money to the farmer round the year.

8) Adoption of New Technology: Resources farmer (big farmer) fully utilizes technology. IFS farmers, linkage of dairy / mushroom / sericulture / vegetable. Money flow round the year gives an inducement to the small/ original farmers to go for the adoption technologies.

9) Saving Energy: To identify an alternative source to reduce our dependence on fossil energy source within short time. Effective recycling technique the organic wastes available in the system can be utilized to generate biogas. Energy crisis can be postponed to the later period.

10) Meeting Fodder crisis: Every piece of land area is effectively utilized. Plantation of perennial legume fodder trees on field borders and also fixing the atmospheric nitrogen. These practices will greatly relieve the problem of non – availability of quality fodder to the animal component linked.

11) Solving Fuel and Timber Crisis: Linking agro- forestry appropriately the production level of fuel and industrial wood can be enhanced without determining

effect on crop. This will also greatly reduce deforestation, preserving our natural ecosystem.

12) Employment Generation: Combining crop with livestock enterprises would increase the labour requirement significantly and would help in reducing the problems of under employment to a great extent IFS provide enough scope to employ family labour round the year.

13) Agro – industries: When one of produce linked in IFS are increased to commercial level there is surplus value adoption leading to development of allied agro – industries.

14) Increasing Input Efficiency: IFS provide good scope to use inputs in different component greater efficiency and benefit cost ratio.

Components in IFS

The components of IFS include crops, fish farming, poultry, pigs, cattle, sheep and goat, fodder production, kitchen gardening etc.

The feeds derived from “alternative” crops require “alternative” farming systems. These are on small-scale and are highly productive. These are diversified and integrated and the role of animals in these systems is synergistic rather than as primary producers. Emphasis is on “small” livestock. External inputs can be minimized through waste recycling, and growing of nitrogen-fixing and pest-resistant plants in the farming system.

IFS models could be developed based on existing production system using concepts and components. The farm has subsystems:

1. Crop production including vegetables
2. Fodder production
3. Pig production
4. Poultry Production (Dual purpose)
5. Cattle production

6. Biogas production

7. Compost/Humus production

Crops

The crop activities in the IFS consist of grain crops (wheat, sorghum, rice, beans and soybeans), vegetable crops, root crops (cassava, cocoyam, and sweet potato), sugarcane, tree crops (moringa, mulberry, leucaena) and fodder crops. The selection of crops is dependent on preferences based on family consumption, market, soil type, and rainfall and type animals raised.

Livestock

The livestock activities in IFS consist of poultry, pigs, cattle and small ruminants. The selection of livestock is also dependent on preference based on family consumption, potential market, and availability of resources.

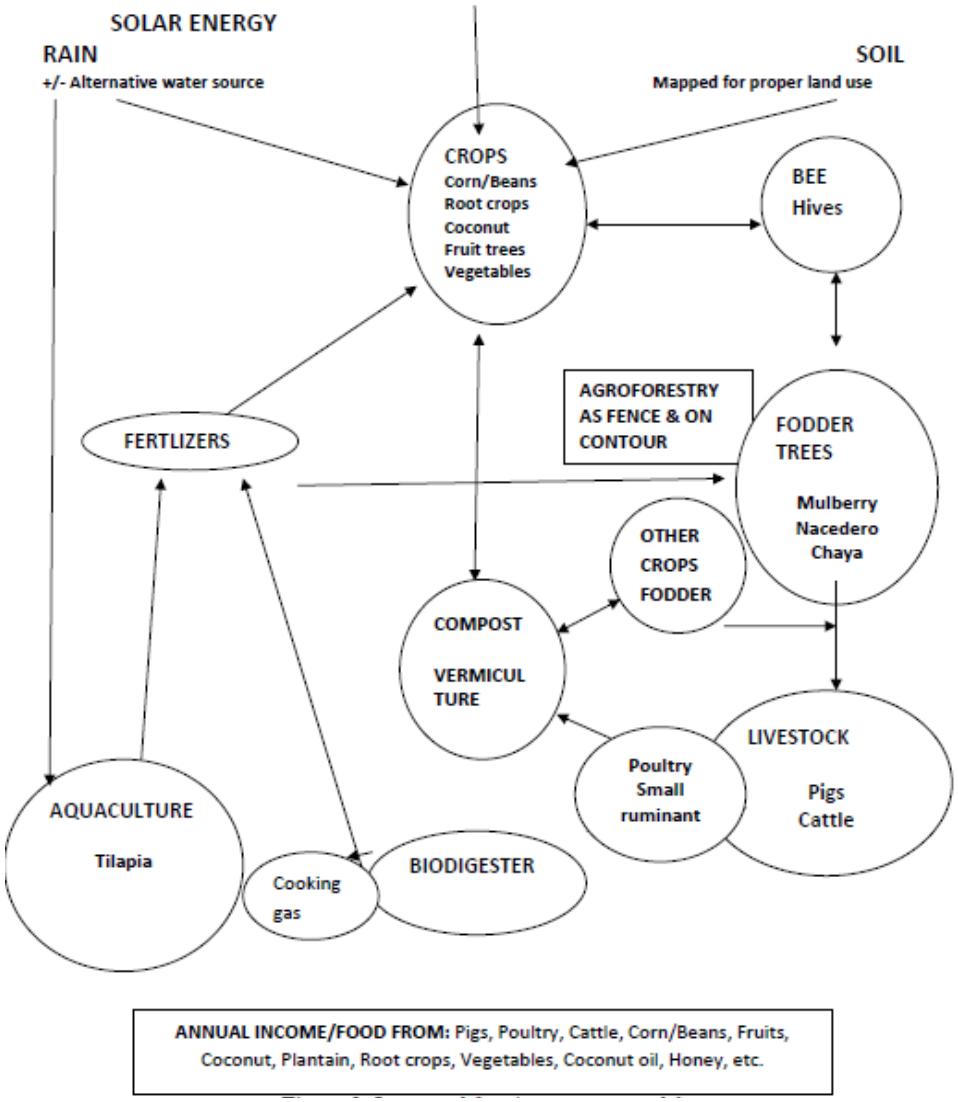


Figure : Integrated farming system model

1. Farming System Under Lowland: Common cropping system in rice based system, especially under agro climatic condition of south India is rice- rice – pulse, modified cropping system includes crops like maize, groundnut, sesamum, rice – fish – poultry culture system appears to more remunerative, poultry droppings from the poultry shed placed well above the farm pond meets the needs of fish in the pond water, Water in the pond can be used for irrigating the crops. About 500 layer

chicks are sufficient and excreta can meet the feed requirement of 7,500 polyculture fingerlings in one ha of pond water rice- fish system is also remunerative.

2. Upland Irrigated Farming Systems: Additional income can be generated by enterprises like dairy biogas and silviculture to the usual cropping systems. Two to three milch cows can be maintained from one ha. Straw recycling of farm and animal wastes through biogas unit can produce cooking gas for family use. Several such integration can greatly increase farm income, besides providing work to family members all through the year.

3. Rainfed Farming Systems: Environmentally sustainable dry land farming systems emphasis conservation and utilization of natural resources. Agronomic practices of conservation, tillage and mulch farming, rotational cropping use of legumes and cover crops for improving soil fertility and suppressing weeds and efficient use of cattle manure are some of the components of sustainable farming systems.

For above regulation sound land use policy is necessary to tackle the problems of deteriorating natural resources, like soil and water. Majority farmers maintaining work animals, milch animal, chicks, sheep, goats etc. with crop production. Farmers approaching farming systems, but benefits from these system is low, because of non- educating and also due to non- adoption of improved technology.

Work has been conducted under AICRP on IFS at various centres and following models have been reported to be economically good (Source, PDFSR, Modipuram)

Model 1. Integrated Farming System (IFS) Model For Haryana

Details of various components of IFS along with allotted area

Treatments	Crop sequence	Area allotted (m^2)
T_1	Cotton – wheat	2000
T_2	Mungbean – wheat (tall)	500
T_3	Pearl millet – mustard	2000
T_4	Sorghum (F) – wheat	2000
T_5	Sorghum (F) – berseem (F)	750
T_6	Sorghum (F) – Oat (F)	750
T_7	Clusterbean-potato-okra	500
T_8	Okra – palak – bottle gourd	500
T_9	Horticultural crop (lemon + guava) + field crop and marigold as intercrop	600
T_{10}	Vermicompost	50
T_{11}	Mushroom unit	50
T_{12}	Dairy unit	300
T_{13}	Boundary plantation with bael	-

The IFS model is being practiced over an area of one ha. The yield and returns varied markedly among different cropping systems. Vegetable component gave net return of Rs 10470/- from an area of 1000 m^2 and cotton-wheat gave net return of Rs. 22608/- from an area of 2000 m^2. Residues of vegetable crops were incorporated in situ whereas cotton sticks were removed from the field. Guava and lemon trees are yet to bear fruits. The income from animal component added from February 2014 is Rs. 14453. The net return from above IFS model was Rs.79672 ha^{-1} $year^{-1}$.

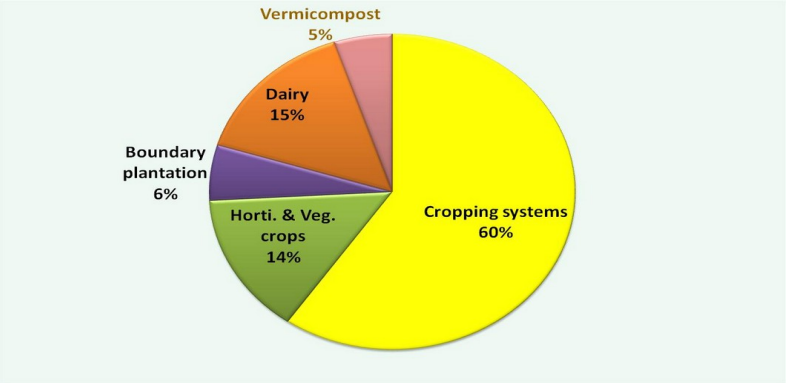


Model 2. Integrated Farming System (IFS) Model for Gujarat

Sr No.	Name of IFS components	Area (ha)	Gross return (Rs/ha)	Cost of Unit (Rs/ha)	Net profit (Rs/ha)
1	Cropping systems	0.700	1,50,955	54,416	96,539 (60%)
2	Horticultural and Vegetables crops	0.250	40,530	17,160	23,370 (14%)
3	Boundary plantation	--	11,801	2620	9,181 (06%)
4	Livestock	0.025	1,11,271	86,886	24,385 (15%)
5	Vermicompost	0.010	16,620	8,310	8,310 (05%)
Total		1.000	3,31,177	1,69,392	1,61,785

Net profit of IFS unit from different components & Percent contribution towards net profit/ha

Total Net Return from 1 ha area Rs. 1,61,785



Model 3. IFS model for marginal farmers under Irrigated Dry situation, Andhra Pradesh

Modules/ Enterprises identified	**Area (ha)**	**Treatments**
Crops	0.70	Cropping Systems:
		Rice-Maize
		Rice-Sunflower
		Maize-Castor
		Pigeonpea +Greengram-Groundnut
		Maize + Pigeonpea - Cowpea (Fodder)
		Hybrid Napier and Lucerne
Horticulture	0.20	
Fruits		Guava
Vegetables -4		Tomato, Green Chillies, Carrot, Cluster bean
Livestock		
Buffaloes -2	0.04	Graded Murrah breed
Goat-5 +1	0.015	Osmanabadi breed
Back yard Poultry		
Fowls (Hens & Cocks)-18 +2		Vanaraja/Rajasri breed
Vermi-compost unit	0.015	
Supporting activities[#]	0.03	
TOTAL	1.00	

Economics of IFS unit

Component	Area (m^2)	Returns (Rs $plot^{-1}$ Yr^{-1})		
		Cost of cultivation	Gross	Net
Crop	6000	31414	57092 (20.92)	25678 (55.76)
Horticulture	2000	9848	8425 (3.09)	-1423 (-3.09)
Livestock	1350	185544	206319 (75.62)	20775 (45.12)
Others (composting-FYM)	650	--	1017 (0.37)	1017 (2.21)
Total	10000	226805	272852	46047

Model 4. Integrated Farming System Model for Jammu

A. Crop Block					
S.No.	**Crop/Area (m^2)**	**Cost of production (Rs)**	**Gross Return (Rs)**	**Net return (Rs)**	**B : C ratio**
1	Basmati-370 (2176)	5392	12960	7568	1.4
2	Rice = IET-1410 (476)	3215	4160	945	0.29
3	Wheat = PBW-550 (2176)	5138	6598	1460	0.28
4	Potato= K. sinduri (576)	4653	6300	1647	0.35
5	Mash = Nul-7 (576)	2738	3190	452	0.16
6	Gobi sarson = GS-1 (1050)	2914	3800	886	0.30
7	Maize = JH-3459 (1050)	3537	4774	1237	0.34
	Overall	27587	41742	14195	0.52

B. Fodder Block					
	Berseem/oat/maize (1298)	3010	10400	7390	2.45

C . Horticulture (3000 m^2)

S.No	**Tree/Number of plants**	**Cost of cultivation**				**Return will realize after 3rd year**
		1st year	**2nd year**	**3rd year**	**Total**	
1	Mango (33)	4473.35	1356.1	1744	7573.45	
2	Guava (118)	3918.35	1429.6	5825	11172.95	
3	Lemon (22)	3223.5	1175	996	5394.5	
4	Aonla (9)	2500	1050	462	4012	
5	Agroforestry (Boundry plantation) (345)	4666.25	-	2800	7466.25	

C: ii Intercropping with vegetable and floriculture in horticulture block

S.No.	**Crop/Area (m^2)**	**Cost of production (Rs)**	**Gross Return (Rs)**	**Net return (Rs)**	**B : C ratio**
1	Brocoli (258)	2717	7000	4283	1.57
2	Cabbage (244)	1046	2600	1554	1.48
3	Knol khol (425)	1189	1675	486	0.40
4	Marigold (1033)	3758	10050	6292	1.67
5	Cauliflower (388)	1100	2900	1800	1.63
6	Raddish (138)	650	1540	890	1.36
	Overall	106460	25765	15305	1.46
D. Animal unit	1 Cow + 1 Buffalo + V.C (1000)	91056	137414	46358	0.51
E. poultry	Broiler + Vanraja	27260	65046	37782	1.39

F. Fishery	1500	4730	10000	5270	1.11
G. Apairy	5 box	3330	5750	2420	0.72
	Overall	**167437**	**296157**	**128720**	**0.78**

Model 5. Relative efficacy of different farm enterprises in the integrated farming system model in Uttranchal

Farm Enterprises	Production (tonnes REY/ha)	Gross returns (Rs / year)	Cost of Production (Rs. / year)	Net returns (Rs. / year)	B:C ratio	Monetary efficiency (Rs/ha/day)
A. Field Crops	9.31	93054 (35.5%)	51708.5	41344.5 (34.8%)	0.80	113.3
B. Agroforestry / Horticulture	8.55	86550 (33.1%)	34950	51600 (43.4%)	1.47	141.4
C. Dairy (1+1) (cows + heifer) + Vermicompost	7.87	82244 (78744+ 3500) (31.4%)	56230	26014 (21.8%)	0.46	71.3
IFS model-Total land (m^2)	25.73	261847	142888.5	118958.5	0.83	325.9

Integrated Farming System (IFS) For Wetland System

1. Cropping (Rice- fish – poultry- mushroom)
2. Fish culture
3. Poultry
4. Mushroom production
5. Cropping 0.36 ha
6. Fish pond 0.04 ha

This farming system was compared with conventional cropping normally followed in the region. Rice – Rice – green gram and rice – rice – green manure (0.20 ha).

Component Description:

1. Fish pond area of 0.04 ha with 1.5 m depth.
2. Diff. finger lings of diff. fish species with total of 7500/ ha.
3. Harvesting of fish commenced from tenth month. A poultry shed at corner of fish pond (shed 2.2 m^2).
4. Free fall of poultry dropping into the fish pond (Twenty Bapcock chicks of 18 weeks old reared).
5. The feed components were purchase only in 1st year of cropping.
6. The birds started laying eggs around 22nd weeks up to 72nd when they were culled out.
7. A mushroom shed of 5 x 3 m was constructed with local materials. Oyster mushroom was produced utilizing rice straw as the base material.

Economics- On an average net profit of Rs. 11,755 was obtained in rice – poultry- mushroom system as compared to Rs. 6, 335 only from conventional system of cropping. Additional employment of 174 man – days was generated due to IFS.

IFS – Tamil Nadu – Cauvery delta Zone (Crop – poultry- fish system).

Farmers of this zone are practicing monocropping of rice for two season followed by a rice fallow pulse. Among the different allied activities, pisciculture plays an important role in this zone since water is available in the canal about 7 – 8 months. Poultry farming is another feasible enterprise. By combing the enterprises of poultry – cum fish culture with rice cropping system the economic status of the small and marginal farmers could be improved.

Components: - One ha area has been selected.

- 0.04 ha area for fishpond
- Improved cropping as rice- rice cotton (0.76 ha) and
- Rice- rice – maize (0.20 ha).

Maize being a major constituent of poultry feed was included in the system. This system compared with the existing practice of rice- rice – black gram

Poultry unit: - 50 bapocock's, Bu 300 hybrid layer bird of 21 weeks age were maintained till 43 Weeks.- 100 g/day/ bird through maize, rice bran, groundnut cake.

Fish culture: Ponds near to poultry shed. Different fingerlings of fish in ponds, with density of population maintained were 10000 fingerlings/ha. The fish were harvested after six and half months.

Economic: A net return of Rs. 17,200 was obtained by integrating different enterprises by introducing poultry – cum – fish culture with cropping a total employment of 385 man- days was generated.

Integrated Farming System (IFS) for Irrigated Situations (Garden lands):

A model integrated farming system to suit the small and marginal farmers of garden land condition was studies at TNAU, Coimbatore, during 1988-1993 (Rangaswami et al. 1995) An area of one ha was selected for IFS and compared with conventional cropping system(CCS).

Components of IFS:

	Cropping	Area (ha)
I.	Cotton = green gram maize + fodder cowpea-Bellary onion	0.56
II	Wheat = sunflower – maize + fodder cowpea-summer cotton + green gram	0.11
III	Grass Bajra Napier ((Co. I)	0.15
IV	Lucerne	0.05
V	150 Terrs of Leucaena (planted in the bunds)	0.05
	Total	1.00

Farm Stead:

Dairy Unit	3 jersey cows + 2 calves
Biogas Unit	2 m 3 capacity
Mushroom Production	1.5- 2.0kg/ha

The above integrate system was compared with the conventional cropping system of cotton – sorghum – finger millet in 0.20 ha area.

Economic returns from the system: Maize flour, cottonseed and wheat bran obtained from the crop components were recycled for preparing dairy feed from the second year. About 45.5 t of to the animals dung was recycled for the biogas plant. Mean revenue of Rs. 34600/ ha was realized in IFS as compared to RS. 13950 obtained in CCS. Employment opportunity was also enhanced to the tune of 770 man- days per year under IFS as against conventional cropping.

III) Integrated farming system (IFS) for Dry Land based system:

Integrated farming system for dry lands suggested for Coimbatore and Aruppukottai, Tamil Nadu, are described below:

Model for Coimbatore, Tamil Nadu (mean annual rainfall: 640 mm).

Crop components for one hectare:

	Cropping	Area (ha)
i)	Sorghum + cowpea both grain purpose	0.20
ii)	Sorghum + cowpea both fodder purpose	0.20
iii)	Leucaena (tree fodder)+ Cenchrus (grass fodder)	0.20
iv)	Aracia Senegal (tree fodder) + grass	0.20
v)	Prosonic cineraris (tree fodder) + grass	0.20

Animal Components: Telicherry goat: 6 (5 female + 1 male – stall fed)
Conventional cropping system = 0.20 ha (sorghum+ cowpea- grain purpose).

The animal components, tell cherry goat, numbering 6 (5 female+ 1 male) were kept in a shed (6 x 4 m). The goals were stall fed form the cropping components. Two kg each of green and dry fodder and 100 g of concentrate were given to each animal. At the end of second year all the 4 male goats were disposed retaining one number. From the end of third year onwards, 20 female and one male were retained and remaining disposed. The litter form the goat shed was used for composting and recycled as manure. A farm pond was dug in an area of 300 m^2 and the rain water harvested was utilized for pot watering the tree saplings.

Economic returns from the system: men additional revenue of Rs. 3750 ha/year was obtained from IFS over CCS. The employment generation under IFS was 153- man days/ ha / year whereas it was only 40- man days/ ha/ year in the CCS.

IFS for island Ecosystem:

Integrated farming system models have been developed for the Andaman and Nikobar Islands –

1.Coconut – cum – fodder- cum – milch cattle: Mixed farming by raising fodder grass coconut has been founded profitable; grass in coconut has been found profitable, grass like hybrid Napier or leguminous fodder like stylo in coconut garden can support 4- 5 dairy animals. Animal supply large qualities of cattle manure applied to coconut garden improve the soil fertility.

2. Coconut – cum fish culture in salt affected lands: Coconut grows successfully even in salt affected paddy fields if the fields are within the approach of brackish water. The bunds of fields at water entrance gate should be raised to maintain the required level of water. Field bunds should also be raised a per the requirement. Water will be exchanged and thus fish raising can be taken up together with coconut in dry period. In rainy season, salinity tolerant rice varieties could be cultivated in the same field with coconut and fish culture.

3. Fruits- fodder- milch cattle: The space available between the fruit trees can be utilized for growing fodder crops such as cow pear, rice bean, germ gram and black gram. The fodder could be used for feeding the cattle and cattle manure could be applied to the fruit trees and fodder crops.

Integrated Farming System for Wet Land

A. IFS

IFS is a resources management strategy to achieve economic and sustained agricultural production through two or more interrelated or interdependent agricultural allied enterprises, to meet diverse requirements

of the farm house hold, while preserving the resource base (Soil fertility) and maintaining a high environmental quality.

Advantages of IFS

• Increased income.

• Income at short periodic intervals from different components/enterprises.

• Yield stability.

• Employment generation throughout the year.

• Effective utilization of farm residues and livestock wastes (better organic recycling).

• Optimal utilization of all resources.

• Meeting the energy need of the family to a considerable extend.

• Production of diversified farm products leading to balanced diet.

• Improved standard of living of the farmers.

B. Wet Land Farming

It is otherwise called as low land farming. Here the soils are usually flooded or copiously irrigated to keep the soil in a continuously submerged condition. Irrigation is through Canals, Ponds or Tanks.

C. Types of IFS

1. Crop based IFS: Here, crop production is the major activity, utilizing a greater share of available resources and often contributing more of total income. Other activities complement the crop production activity by way of recycling of organic waste and supplementing the income. The crop by products can be utilized for the allied enterprises. For example,

Crop + Dairy

Crop + Poultry

Crop + Fish

2. Live stock based IFS: Here, rearing of livestock is the major activity, which contributes the major share of income. Cropping is to plan in such a way to meet the fodder and feed requirement of the livestock. For example,

Poultry Farm

Goat/sheep Farm

Fish + Piggery + Crop

3. Tree based IFS: Cultivation of multipurpose tree crop to meet the requirements of fuel, feed, fodder, wood and timber. Annual field crops will be raised as intercrop in between tree crops.

4. Horticulture based IFS: Growing of either vegetable crops and fruit trees or both serve as the major component. In between fruit trees, annual field crops can be cultivated.

D. Features of Wetland

Cropping: 9–12 months in a year.

Source of Water: Canals, Ponds, Tanks.

Climate: Arid and Semiarid.

Problem: Over irrigated/water logging.

Fertilizer application: Liberal to maximize the yield.

Other problems: Drainage, soil health due to continuous submergence, salt affected soils, mainly surface salinity.

E. Crops in Wetland

Rice–rice–rice with assured irrigation

Rice–rice–rice fallow pulses with limited use of water

Fallow–rice–rice fallow pulses with limited use of water

Rice–rice–upland crop (ground nut)

Upland crop–rice–rice

Rice–rice–cotton

F. IFS for Wetlands

IFS approach introduces a change in the farming techniques for maximum production in the cropping pattern and takes care of optimal utilization of resources in which farm wastes are better recycled for productive purposes. It comprises the cropping system, the livestock system and the farmhouse hold.

The final outcome is to get into the useful products that can be consumed or sold. Selection of suitable agricultural enterprises, suited to the given agro-climatic conditions and socio–economic status of the farmers would bring prosperity in the farming.

Components/enterprises proposed for IFS under wetland conditions

• Multiple cropping including sequential cropping and multi-tier cropping inter and relay cropping.

• Integrating livestock, cow, goat, piggery etc., along with cropping enterprises.

• Combining fish (Aquaculture) cum poultry in the existing system.

• Agro-forestry on tree components in the existing system.

• Inclusion of mulberry cultivation and sericulture or lac culture with crops.

• Mushroom cultivation with cropping.

• Inclusion of apiary and horticultural crops etc.

I. IFS Models for Coastal Ecosystem

i. Crop: rice

Early duration – Rohini, Cauvery, SSRC 91216 culture

Medium duration – IR 20, TRY-1

Long duration – Pankaj, Jaganath

Flood resistant – IR43, ADT40, TRY-1.

ii. Livestock system

Cows, Poultry, Goat, Fish, Prawn and Duck.

Prawn : Water canals in homestead or coconut groves connected directly or indirectly with backwater having water having free tidal water movements can be converted into productive prawn farms. From an area of 400 m^2 of water canal, 12–16 kg of prawn (Indian white variety) can be produced in 90 days.

It can fetch Rs. 50–60 per kg. The main expenditure will be only Rs. 2000/- for providing sluice gates to regulate the flow of water. Prawn culture can be integrated with duckery, poultry, agricultural, horticulture or piggery.

Fishes: In fresh water areas, fishes of Indian major carps *viz*., roghu, catla and mirgal are grown in perimeter trenches in rice plots.

Tree Crops: Major tree crops suitable for coastal region are coconut, cashew nut and jackfruit. The forest tree species are *Casuarina sp., Acacia sp., Eucalyptus terticornis, Pongamia pinnata, Azadirachta indica and Anacardium* sp.

Farm Pond: The farm pond technology which involves digging of a pond in 20% of the area in coastal lowland area and spreading the excavated earth in remaining 80% of the field is gaining popularity among the farmers in coastal region. In the field so developed, the farmers are able to get a samba crop of high yield rice variety in monsoon season and vegetable in winter and summer months.

II. IFS Model for single crop wetland area (Periyar-Vaigai-command area),

Tamil Nadu, India

Crop + vegetables + poultry + fish

Cropping (in ha)

Rice – Maize 0.20

Rice – Green gram 0.20

Rice – Groundnut 0.20

Rice – Green manure 0.20

Vegetables

Bhindi – Tomato–Brinjal 0.05

Bitter gourd – Moringa–chillies

Poultry unit

Layer – 20 nos.

Fishery unit

Mushroom cultivation if condition prevails.

III. IFS model for a double crop wetland

Components : Cropping–rice

Fish

Poultry

Mush room

Area Allocation

(a) Cropping (in ha)

Rice–Rice–Maize 0.16

Rice–Rice–Ground nut 0.10

Rice–Rice–Sesamum 0.10

Fish pond 0.04

Total area 0.40

(b) Fishery

(400 fingerlings) 0.04

(c) Poultry unit

(over fish pond) 20 layers

(droppings as fish feed)

(d) Mushroom Production capacity of 2.0 kg/day/unit

Integrated Farming System for Irrigated Upland

It is a type of farming where the crops are grown with supplemental irrigation by using water from underground sources. This can also be called irrigated dry land.

A. Features

• Farming will be for around 9-12 months.

• The sources of water are wells and deep bore wells in addition to rainfall.

• The climate is arid to humid.

• Water management is the main criteria, *i.e.*, economic use of water.

• Fertilizer use–liberal to maximize the yield.

B. Constraints

• Salt affected soils are more

• Energy for lift irrigation is a must, but availability is a problem

C. Crops

Crops that can tolerate mild winter

Legumes – green gram, black gram, soybean, and groundnut.

Cereals – maize, sorghum, pearl millet and ragi.

Other crops – cotton, sunflower and vegetables.

Crops that can tolerate cool winter are wheat, barely, mustard and potato.

I. Integrated farming system model for irrigated upland system

(Crop + Dairy + Biogas + Mushroom)

Crop Management Area (ha)

1. Cotton + green gram – Maize 0.56

2. Soybean – Maize-Cotton 0.19

3. Cumbu Napier grass 0.15

4. Lucerne 0.05

5. *Leucanea leucocephala* 150 trees (border trees)

6. Farm shed

Dairy Component – 3 Jersey cows + 2 calves

Biogas – For effective recycling of farm and animal waste, a biogas unit of 2 m^3 capacity

Spawn and mushroom production – 1.5–2.0 kg/days

Integrated Farming System for Dry Land

Crop production depends entirely on the quantity of rainfall received or with conserved moisture. Hence, there is every likelihood of crop being facing with mild to severe stress during its growth period. The major constraint is the wind and water erosion.

Crops

Cereal + Pigeon Pea

Cereal + Cotton

Single crop area - Sorghum, Chickpea, Sunflower, Coriander

Double crop area - Sorghum/Pearl millet/Pulse during rainy period.

Sunflower/Safflower/Chickpea during post rainy period.

I. IFS model for dry lands of Coimbatore (low rainfall < 700 mm), Tamil Nadu, India

Area (ha)

1. Sorghum + Cowpea (grain) - 0.2
2. Sorghum + Cowpea (fodder) - 0.2
3. *Leucanea* + *Cencherus ciliaris* - 0.2
4. *Acacia senegal* + *Cenchrus ciliaris* - 0.2
5. *Prosophis Cineraria* + *Cenchrus ciliaris* - 0.2

Tellicherry goats (20 does + 1 buck)

Chapter 11. Importance and role of IFS in Organic Farming

Organic agriculture has grown out of the conscious efforts by inspired people to create the best possible relationship between the earth and men. Since its beginning the sphere surrounding organic agriculture has become considerably more complex. A major challenge today is certainly its entry into the policy making arena, its entry into anonymous global market and the transformation of organic products into commodities. During the last two decades, there has also been a significant sensitization of the global community towards environmental preservation and assuring of food quality. Ardent promoters of organic farming consider that it can meet both these demands and become the means for complete development of rural areas. After almost a century of neglect, organic agriculture is now finding place in the mainstream of development and shows great promise commercially, socially and environmentally. While there is continuum of thought from earlier days to the present, the modern organic movement is radically different from its original form. It now has environmental sustainability and productivity at its core, in addition to the founders concerns for healthy soil, healthy food and healthy people.

Concept of organic farming

Organic farming is very much native to India. This concept of organic farming is based on following principles:

- Nature is the best role model for farming, since it does not use any inputs nor demand unreasonable quantities of water.

- The entire system is based on intimate understanding of nature's ways of replenishment. The system does not believe in mining of the soil of its nutrients and do not degrade it in any way.
- The soil in this system is considered as a living entity
- The soil's living population of microbes and other organisms are significant contributors to its fertility on a sustained basis and must be protected and nurtured, at all cost.
- The total environment of the soil, from soil structure to soil cover is more important and must be preserved.

Organic agriculture perspective under Indian conditions

Organic farming denotes a holistic system of farming which optimizes productivity in a sustainable manner through creation of interdependent agri-eco systems where annual crop plants (e.g. wheat), perennial trees (e.g. horticulture) and animals (including fishes where relevant) are integrated on a given field or property. It is often confused with certified organic farming or labeled organic produce, which is a commercial quality control and marketing mechanism entailing third party certification or any other form of independent certification for individual commodities and the process of their cultivation. It is important to separate the two.

The first - organic farming (OF) is essentially an agricultural activity employing the knowledge/understanding of naturally occurring processes while the second – a certified organic farming (COF) is essentially the same as OF but there is an addition of checks and balances to maintain soil fertility and control insect-pests and diseases. It is done by involving record keeping on various aspects and is dictated by market forces.

About 74% farmers in India are small and marginal farmers. Organic farming is most relevant to them because they are resource poor to provide costly inputs for enhancing yield. Their only resource viz. land, need to be

prevented from degradations. In the organic farming system approach, a piece of land is used optimally and to its fullest potential to produce a range of nutritious and healthy food as well as other required commodities in a manner which can healthily feed a small family, and maintain soil health and productivity by agricultural practices based on principles of nature. Pests (both insects and diseases) are also controlled and managed by the selection of crop mixes and using biological control measures.

Emergence of organic farming in India

Organic farming has made a credible advancement during the past decade. It is a combined effect of farmers' efforts, NGOs work, Governmental interventions and market forces that organic farming in India has reached a stage where it can swiftly move to occupy prominent space in Indian agriculture. Certified organic farming, which was 42,000 ha during 2003-04, has now grown almost by 20 fold during the last 4 years. As on March 2009, the total cultivated area under organic certification process has crossed one million ha mark. Besides this, there is another 2.5 million ha under certified wild forest harvest collection area. National Project on Organic Farming (NPOF) and National Horticulture Mission (NHM) scheme of Department of Agriculture and Cooperation has significantly contributed to this growth. For quality assurance the country has internationally acclaimed certification process in place for export, import and domestic markets. The National Programme on Organic Production (NPOP) provides necessary policy support. Presently,16 accredited certification agencies are looking after the requirement of certification process and the products certified by them are accepted in many countries including European Union and USA.

Definition

Organic farming "is a production system which avoids or largely excludes the use of synthetically compounded fertilizers, pesticides, growth regulators, and livestock feed additives. To the maximum extent feasible, organic agriculture systems rely upon crop rotations, crop residues, animal manure, legumes, green manure, off-farm organic wastes, mechanical cultivation, mineral bearing rocks, and aspects of biological pest control to maintain soil productivity, tilth, to supply plant nutrients, and to control insects, weeds, and other pests" (USDA, 1980).

The concept of the soil as a living system which must be "fed" in a way that does not restrict the activities of beneficial organisms necessary for recycling nutrients and producing humus is central to this definition.

"Organic agriculture is a holistic production management system which promotes and enhances agro-ecosystem health, including bio-diversity, biological cycles and soil biological activity. It emphasizes the use of management practices in preference to the use of off-farm inputs, taking into account that regional conditions require locally adapted systems. This is accomplished by using wherever possible, agronomic, biological, and mechanical methods, as opposed to using synthetic materials, to fulfil any specific function within the system.

Principles of organic farming

1. To produce food of high nutritional quality in sufficient quantity.
2. To interact in a constructive and life enhancing way with all natural systems and cycles.
3. To encourage and biological cycles within the farming system, involving micro-organisms, soil flora and fauna, plants and animals and careful mechanical intervention.

4. To maintain and increase long-term fertility of soils.
5. To promote the healthy use and proper care of water, water resources and all life therein.
6. To help in the conservation of soil and water.
7. To use, as far as is possible, renewable resources in locally organized agricultural systems.
8. To work, as far as possible, within a closed system with regard to organic matter and nutrient elements.
9. To work, as far as possible, with materials and substances which can be reused or recycled, either on the farm or elsewhere?
10. To give all livestock conditions of life which allow them to perform the basic aspects of their innate behaviour.
11. To maintain all forms of pollution that may result from agricultural practices.
12. To maintain the genetic diversity of the production system and its surroundings including the protection of wild life habitats.
13. To allow everyone involved in organic production and processing a quality of life confirming to the UN Human Rights Charter, to cover their basic needs and obtain an adequate return and satisfaction from their work, including a safe working environment.
14. To consider the wider social and ecological impact of the farming system.
15. To produce non-food products from renewable resources, which are fully degradable.
16. Weed, disease and pest control relaying primarily on crop rotation, natural predators, diversity, organic manuring, resistant varieties, and limited (preferably minimal) thermal, biological and chemical intervention.

17. To create harmonious balance between crop production and animal husbandry.
18. To encourage organic agriculture associations to function along democratic lines and the principle of division of powers.
19. To progress towards an entire production, processing and distribution chain which is both socially just and ecologically responsible.

Relevance of organic farming

Interest in organic agricultural methods is growing, especially in areas where the present modern farming system has unleashed many agro-ecological and environmental problems both on and off the farm, which threaten food security. The following are some examples:

a) Degradation of soil quality (structured & fertility)

b) Pollution of soil, water and food with pesticides and nitrates

c) Health effects on farmers, farm workers, farm families, rural communities (apart from concerns about the non-intended effects of pesticides on human beings in general, sound use of pesticides requires a technical knowledge which is often lacking in developing countries)

d) Resistance of pests to pesticides

e) Dependence on off-farm agricultural inputs which can increase poor farmers' dependence on credit facilities (to purchase synthetic fertilizers, pesticides and seed), which may result in decreased local food security and self reliance. Further consumer awareness of the environmental costs of agriculture is increasing. The awareness of environmental quality and health is often promoted by environmental groups, especially in developed countries. The resulting demand for organic products creates the opportunity to sell organic products at premium prices, enabling organic farmers to continue, and often expand. Some governments have

begun to recognize the possibility that it may be cheaper to support organic agriculture than to rectify problems associated with certain resource- destruction production practices. For this reason, several governments have introduced subsidies for organic agriculture. For example, in Indonesia where, after a period of subsidies on pesticides, the use of this input was prohibited while efforts were put in IPM programmes. In China, pesticide problems in products both on the domestic and export market has resulted in government involvement in certification organizations for “green food”, including also a small amount of organic produce. Both these policies facilitate a shift towards organic agriculture.

Relative characteristics of Modern and Organic Farming systems

Production factor	**Modern**	**Organic**
Productivity	High	High
Sustainability	Low	High
Farming system	Simple	Complex
Bio-diversity	Uniform	Diverse
Production orientation	Market	Subsistence/market
Usage of external inputs	High	Low
Fertilization	Inorganic	Organic
Plant protection	Curative & inorganic	Preventive & organic
Energy balance	Negative	Positive

Components of organic farming

Thus organic agriculture is comparatively free from the complex problems identified with modern agriculture. It is basically a farming

system, devoid of chemical inputs, in which the biological potential of the soil and the underground water resources are conserved and protected from the natural and human induced degradation or depletion by adopting suitable cropping models including agro-forestry and methods of organic replenishment, besides natural and biological means of pest and disease management, by which both the soil life and beneficial interactions are also stimulated and sustained so that the system achieves self regulation and stability as well as capacity to produce agricultural outputs at levels which are profitable, enduring over time and consistent with the carrying capacity of the managed agro-ecosystem.

Crop production and health in organic farming systems is attained through a combination of structural factors and tactical management components to ensure products of sufficient quality and quantity for human and livestock consumption.

1. Diverse crop rotations: Crop diversification can deliver many agronomic and ecological benefits simultaneously, while maintaining or enhancing the scale and efficiency of production. Benefits of diverse crop rotations include yield stability, reduction in disease incidence & severity, reduced pest incidence, improved weed control, reduced soil erosion, recycling of nutrient reserves, transfer or nitrogen from nitrogen fixing species, structural improvement etc. There are many different forms of crop diversification viz., rotational cropping, sequential cropping, intercropping, multi-storeyed cropping system etc., and in practice these can be combined within the farming system. Crop and variety choice and their spatial and temporal design are critical in ensuring an effective rotation. The inclusion of crops, which are able to fix atmospheric through symbiotic relationship with N-fixing bacteria that nodulate on crop roots, enables organic farming systems to be self sufficient in nitrogen.

2. Soil fertility management: The aim of nutrient or soil fertility management within organic farming systems is to work, as far as possible, with in a closed system .Organic farming aims to manage soil fertility through use of organic manures (FYM & farm compost, vermicompost), recycling of crop residues such as straw, plant residues, grasses etc., dung and urine from domesticated animals and wastes from slaughter houses, human excreta & sewage, biomass of weeds, organic wastes from fruit and vegetable production & processing units and household wastes, sugarcane trash, oil cakes, press mud and fly ash from thermal power plant. Biological nitrogen fixation through blue green algae, azolla for rice, rhizobium for legumes, azatobactor & azospirillum for other crops, green manuring & green-leaf manuring, manure form biogas plants, legumes in crop rotations & intercropping systems.

3. Weed control: Organic farmers often identify weeds as their key problem. Within organic systems an integrated approach to weed control using a combination of cultural and direct techniques is necessary. Appropriate soil cultivation viz., deep ploughing in summer, harrowing, inter-cultivation using mechanical hoes and harrows, and the timing of field operations and good crop establishment are vital for successful control of weeds. Mulching the soil surface can physically suppress weed seedling emergence. Soil solarization, to heat field soil under plastic sheeting to temperatures high enough to kill weed seeds (>65°C) can also be used for weed, control in some parts of India. Good seedbed preparation, timely sowing, line sowing, crop rotation, smoother crops & intercropping systems etc., suppress the weed growth and favour normal growth and development of crops in organic systems.

4. Natural pest and disease control: One of the important features of organic farming is the exclusion of plant protection chemicals for pest and

disease control. The system relies on the on-farm diversity, improved health of the soil and crops, protective influence of beneficial soil organisms against soil borne pathogens and use of plant based insecticides and biological control measures. The population of naturally occurring beneficial insects and other organisms which act as bio control agents multiplies making natural control of pests possible when the system is free from the indiscriminate use of chemicals.

Few examples are:

a) Manipulation of crop rotations, to minimize survival of crop specific pests (in the form of, for example insect eggs, fungi) which can infest the next crop

b) Strip cropping, to moderate spreading of pests over large areas

c) Manipulation of the moisture level or pH level of the soil (in irrigated areas)

d) Manipulation of planting dates, to plant at a time most optimal for the crop, or least beneficial for the pest

e) Adjustment of seeding rate, to achieve an optimal density given the need to check weeds or avoid insects

f) Use of appropriate plant varieties for local conditions

g) Biological control methods, to encourage natural enemies of pests by providing habitat or by breeding and releasing them in areas where they are required.

• *Bacillus thuringensis* against caterpillars of *Heliothis, Earias, Spodoptera* etc

• *Pseudomonas fluoroscenes* against *Pythium spp., Rhizoctonia spp., Fusarium spp.*

• Nematodes like Green commandoes and Soil commandoes against caterpillars & grubs

• Nuclear Polyhedrosis virus (NPV) against caterpillars

• *Trichoderma virdi* against many common diseases of vegetables and spices

• Weevils *Neochitina eichorniae* & *N. bruchi* against water hyacinth

• Beetle *Zygogramma biocolorata* against parthenium

h) Trapping insects, possibly with the use of lures such as pheromones

i) Use of domesticated birds

j) Biological pesticides (for example neem oil, nicotine) of which the active ingredient is short-lasting, and which may be produced locally

5. Integrated nutrient management

Integrated nutrient management system envisages conjunctive use of organic manures, crop residues, biofertilizers, legumes in crop rotation and green manuring. It combines traditional and improved technologies to gain from the symbiosis and synergy of crop- soil-environment bio-interactions. The concept is for optimization of all available sources of plant nutrients to improve soil fertility availing nature's gifts.

Development of INM system involving and appropriate mix of organics, biological N fixation, phosphate solubilising microbes, and need based chemical fertilizers would be crucial for sustainability of production and soil as a resource base for it.

Bulky organic manures: In India, the estimate production of rural compost is about 226 million tons and urban compost of 6.6 million tons annually. Aggregate stability, decrease in pH, resistance to compaction, infiltration and water holding capacity. Proper methods of preparation of FYM/Compost therefore have to be popularized.

Recycling of organic wastes: Substantial quantities of crop residues (350 million tons) are produced in India every year. Crop residues in combination with organics have been shown to improve availability of plant nutrients,

soil organic matter, aggregate stability, infiltration rate, microbial population etc.

Bio-fertilizers: Bio-fertilizers such as rhizobium culture are an effective source of N supply to leguminous corps. Azotobacter and Azospirillium help in N fixation and supply to crops like rice, wheat, sorghum, maize, cotton, sugarcane, fruit corps and vegetables. Phosphate solubilising bacteria viz., *Bacillus aspergillus* helps in making available soil P to the crops and increases the solubility of indigenous sources of P like rock phosphate. Blue green algae and Azolla have shown promise in low land rice. These are renewable and environment friendly supplementary sources of nutrients and are presently being used in quantities between 8-10 tons per year. *Vesicular arbuscular mycorrhiza* (VAM) has beneficial effect on plant growth, particularly in P deficient soils. Improved uptake of water, production of plant hormones and microbial activity are the prime benefits of mycorrhizal inoculations.

Green manuring: Green manuring is a cheap alternative to the use of fertilizer N. The process also makes a positive contribution to the maintenance of soil organic matter at a satisfactory level. The stem nodulating green manure plant, *Sesbania rostrata* (Dhaincha) can fix 100-250 kg/N/ha in 45 to 55 days and has great scope in rice culture. There is also greater scope for green-leaf manuring for rice and other crops from the lopping of various multipurpose trees popularized through afforestation and agro-forestry systems.

Popularization of bio-gas plants, encouraging legumes in crop rotation and intercropping system and use of sewage, sludge and effluents for agriculture can also be the components of INMS.

Chapter 12. Low input sustainable agriculture and low cost agricultural technologies

Low Input Sustainable Agriculture (LISA)

It is a form of agriculture wherein the use of external inputs including chemical inputs is minimized and production is kept sustainable. It uses synthetic fertilizers or chemical pesticides below rates commonly recommended by the Extension workers. There is no elimination of these materials. Any system that reduces purchased chemical inputs can be called low-input farming. Yields are maintained through greater emphasis on crop rotation, crop residues, animal manures, bio-fertilizers, IPM, IWM and utilization of on-farm resources and management. Locally available resources are used by maximizing the complementary and synergistic effects of different components of the farming systems. There is use of External inputs in a complementary way. Main advantages of this type of agriculture are:-

1. Ecological balance
2. Low cost of production
3. Healthy environment
4. Overall risk of the farmer is considerably reduced
5. Ensure both short and long term profitability
6. Food very little or no pesticide residue is ensured
7. High diversity, renewable and biodegradable inputs are used
8. Rate of extraction of resources do not exceed rate of regeneration

Some of the disadvantages may be:-

1. Lower yield
2. Ineffective control of weeds, insect-pests and diseases.

IPM is probably the oldest and widely accepted Extension Service programme devoted to low-input agriculture. However, only recently have the “non-chemical” approaches-such as cultural, mechanical, and biological-within the IPM framework been emphasized over the chemical component. Such programmes are now called *biologically-intensive IPM.*

LISA systems are mainly based on preventive approach whereby the problem is tackled at its roots, as compared to symptom-curing nature of HEIA. Ecological and biological principles are the basis of the farm system. Nature works for the farmer, the farmer does not have to work against it. They are often based on ITK and production systems adapted to modern requirements and much less on external expertise.

- Supports profitable production
- Protects environmental quality
- Uses natural resources efficiently
- Provides consumers with affordable, high-quality products
- Decreases dependency on non renewable resources
- Enhances the quality of life for farmers and rural communities and
- Will last for generations to come.

Criteria for LISA

Ecological criteria

1. Balanced application of plant nutrients
2. Judicious use of water
3. Diversity of genetic resources
4. Efficient utilization of genetic resources
5. Efficient utilization of energy sources
6. Reducing harmful impact on environment
7. Reducing dependence on external inputs

Economic criteria

1. Sustained farmer livelihood systems
2. Competitiveness
3. Efficient utilization of production factors
4. Reduced relative value of external inputs

Social criteria

1. Wide-spread and equitable adoption potential, especially among small farmers
2. Reduced dependency on external institutions
3. Enhanced food security at the family and national level
4. Respecting and building on indigenous knowledge, beliefs and value systems
5. Contribution to employment generation

LISA is a form of agriculture that maintains productivity for a long period by:

- Increasing the use of locally available resources which complement each other and thus have synergistic effects
- Reducing the use of off-farm external and non-renewable inputs with potential of degrading environment
- Improving the match between cropping patterns, productive potential and environmental constraints for longer sustainability of the system
- Making optimum use of biological and genetic potential of plant and animal species
- Taking full advantage of ITK and practices, innovative approaches not fully understood by scientists

Low cost technology and non- monetary inputs

The technology that does not need high cost- or – more investment is called Low cost technology.

1. Cultivar selection: Numbers of high yielder cultivar of different crops, resistant / tolerate to pest and disease have been developed.

2. Time of sowing: In time seed bed preparation and timely sowings as per the season. (Kharif/ Rabi/ summer) for any crop, results in even and better germination, well establishment and their by minimize the crop loss.

3. Plant population: keeping the total plant population constant, inter and intra row plant population can be adjusted to minimize the humidity buildup within the crop canopy.

4. Timely intercultivation: Manual or mechanical inter- cultivation suppresses the pest, diseases and weeds similar to that of preparatory tillage. Weeds that could be serving as alternate host to insect and pathogen can be effectively controlled by inter- cultivation.

5. Crop rotation: Helps as complementary effect, that legumes inclusion in system add the nitrogen and residue, thereby increases the water holding capacity of soil as well as reduction in infestation of weeds which are associated with crop and minimize the buildup of pest and diseases.

6. Strip cropping: inclusion of erosion permitting and erosion resisting crop help in conservation of moisture and increases in water intake.

Suggested Reading

1. A text book of Agronomy by B. Chandrasekaran, K.Annadurai and E.Somasundaram.

2. Cropping and farming systems by S.C.Panda

3. Principles and practices of agronomy by P. Balasubramaniyan and S.P.Planiappan

4. Organic farming- theory and practice by S.P. Planiappan and K.Anandurai

5. Sustaining our food security by R.S.Paroda, Konark Pub.

References

Adetiloye, P.O. and Ezedinma, F.O.C., 1983. Land equivalent coefficient concept for the evaluation of competitive and productive interactions in simple to complex crop mixture. Ecological Modeling, **19**: 27-39.

Azam Ali, S. N., Mathews, R.B. and Williams, J.H., 1990. Light use, water uptake and performance of individual components of sorghum/groundnut intercrop. Experimental agriculture, **26**: 413-427.

Balasubramanian, V and Sekayange, S, 1990. Area harvests equivalency ratio for measuring efficiency in multiseason intercropping. Agronomy Journal, **82**: 512-522.

Chung, F.T., 1973. Analysis of change of Faiwan's cultivated land utilization for recent years. Rural economic division, JCRR report 21, Faipei, Taiwan.

Darlymple, D.F., 1971. Survey of multiple cropping in less developed nations. USDFA, Washington, DC.

Jayanthi, C., 2002. Farming System – principles and practices.

Lal, R. and Miller, F.P., 1990. Sustainable farming systems for tropics. International symposium on natural resource management for a sustainable agriculture, Indian Society of Agronomy, New Delhi.

Menegay, M.R., Hubbel, J.N. and William, R.D., 1978. Crop intensity index: A research method for measuring land use in multiple cropping. Horticultural Science, **61**: 440-442.

Rangaswamy, A., Venkataswamy, R. and Palanniappan, S. P. 1996. Rice-poultry- fish-mushroom integrated farming systems for Tamilnadu. Indian Journal of Agronomy. 41(3): 344-348.

Singh, A., 1972. Conceptual and experimental basis of cropping patterns. *In:* proceedings of symposium on cropping patterns in India. ICAR, New Delhi.

Strout, A.M., 1975. Some definitional problems with multiple cropping diversification. Philippine Economist, **14**: 308-316.

Tondon, H.L.S., 1991. Sulphur research and agricultural production in India. FDCO, New Delhi.

Willey, R.W., 1979. Intercropping- its importance and research needs 1. Competition and yield advantage and 2. Agronomy and research approaches. Field Crop Abstracts, **32**: 1-10, 73-85.

Willey, R.W., Rao, M.R. and Natarajan, M., 1981. Traditional cropping systems with pigeonpea and their improvement. In: proceedings of the international workshop on pigeonpea. ICRISAT, Hyderabad, India.

PRACTICAL

Exercise No.1. Preparation of cropping system for different farming situations having varying resource availability

Promising Rice based cropping systems in India

System	Region
Rice - Indian mustard	Kashmir valley
Rice - wheat - sorghum (Fodder)	Jammu region
Rice-wheat- green gram	Punjab, Haryana, U.P.
Rice- Mustard	North Konkan and Assam
Rice - field bean	South Gujarat
Rice - groundnut	South Karnataka
Rice - black gram/sesame/cotton	Thanjavur, Tamil Nadu (Old Cauvery delta)
Rice - groundnut	Thanjavur, Tamil Nadu (New Cauvery delta)
Rice - wheat - jute	coastal A.P., Orissa
Rice - green gram	West Bengal
	Eastern Madhya Pradesh

Crop sequences and their production potential

Crop sequences	Cropping pattern	Total production (q/ha)
Two crops	Maize — wheat	90.1
Three crops	Green gram - maize - wheat Green gram - pigeon pea - wheat Cowpea - pearl millet —wheat	97.7 68.8 92.9
Four crops	Green gram - maize - potato -wheat Green gram - maize - toria -wheat	135.9 112.8

Potential cropping systems for different agro climatic situations

Region	**System**
Western Haryana	Cotton - wheat
Rajasthan	Groundnut - wheat
Madhya Pradesh	Soybean - wheat
North Gujarat	Pearl millet - mustard
Saurastra region - Gujarat	Cotton - groundnut
Western Maharashtra	Sorghum - wheat
Western Maharashtra	Groundnut - wheat
Marathwada region of Vidarba	Cotton - groundnut
North Karnataka	Sorghum - bengalgram
South Karnataka	Cotton - groundnut

Cropping pattern with varying rainfall and soil moisture storage capacity

Rainfall (mm)	Storage capacity of soil (mm)	Cropping pattern
350 — 625	100	Single crop in *Kharif*
650 — 750	100	lntercropping canjpe attempted
780 — 900	150	Sequential cropping is possible
900 - above	200	Sequential cropping is assured

Cropping system for different region of India under dry lands of Northern region

Regions/soil type	Crops	
	Kharif	*Rabi*
Samba (Jammu)	Maize	Wheat
	Green gram	Barley
	Cowpea	Wheat
	Sunflower	Wheat / barley
	Groundnut	Barley
	Pearl millet	Bengal gram
Punjab	Sorghum	Wheat
	Maize	Wheat/potato/barley

Hisar (Haryana)	Pearl millet	Chickpea
Arid soils	Mono-cropping	

Besides these examples of cropping systems for different farming situations having varying resource availability will also be discussed.

Exercise No. 2. Working out input requirement and preparation of calendar of operation

Input requirement:

1. Find out the seed rate per ha of cotton in kg/ha with the help of following observations:

a. Spacing= 75x30 cm

b. 1000 seed weight of fuzzed seeds=140 gms and unfuzzed seeds= 150 g

c. Purity % of the seeds=95

(Ans: 7.27 kg fuzzed and 7.8 kg unfuzzed seeds)

2. Estimate the seed rate per ha for wheat crop with the help of following data:

a. Row spacing=20 cm

b. Av. spacing between plants= 2.5 cm

c. Germination of seeds= 95%

d. Purity of seed=95 %

e. 1000-seed weight=46 g

(Ans: 102 kg)

3. Calculate quantities of CAN, DAP and murate of potash for a crop to be grown in a plot size of 50x20 m. The crop requires 120 kg N, 60 kg P_2O_5 and 60 kg K_2O per ha. The plot was supplied with FYM at @ of 10 t/ha.
(Ans. 7.6 kg DAP, 22.5 kg CAN and 2.1 kg MOP).

Calender of operations for different enterprises/components in IFS will be discussed in class. An example of wheat crop is given here for students:

Seed bed preparation

The seed bed is prepared for sowing of wheat after pre-sowing irrigation. The field is harrowed twice, two ploughing with cultivator followed by one planking in order to have a good tilth with sufficient moisture to ensure proper germination.

Fertilizer application

All P (kg ha^{-1} as DAP), K (kg ha^{-1} as MOP), Zn (kg ha^{-1} as zinc sulphate) and half of N (kg ha^{-1} as urea) is applied as basal dose and half N is top dressed after first irrigation.

Sowing

The seeds are sown with seed drill @ 125 kg ha^{-1} keeping inter row distance of 20 cm.

Irrigation application

Post sowing irrigations are applied using the surface method of irrigation. Six irrigations are required for wheat crop.

Hoeing and weeding

For control of grassy as well as broad leaved weeds one spray of recommended herbicide done after CRI irrigation. Two hand weeding at 35 and 60 DAS can be done.

Harvesting and threshing

Harvesting is done with the help of sickles. The bundles of wheat crop are sun dried and then weighed for recording straw yield. Grain yield is recorded after threshing and later converted into kg ha^{-1}.

Schedule of cultural operations in wheat crop.

Sr. No.	Nature of operations	Date of operation (To be completed by the students in the class)
A. Pre-sowing operations		
1.	Pre-sowing irrigation	
2.	Field ploughing and seed bed preparation	
3.	Layout, fertilizer application and sowing	
B. Post sowing operations		
4.	Final layout	
5.	First irrigation	
6.	Fertilizer application	
7.	Weedicide application	
8.	Hand weeding (first)	
9.	Second irrigation	
10.	Hand weeding (second)	
11.	Third irrigation	
12.	Fourth irrigation	
13.	Harvesting	
14.	Threshing	

Like the example of input requirement and calendar of operations in wheat, the students will be taught the example for other enterprises of IFS like dairy, poultry, bee keeping, mushroom etc.

Exercise No. 3. Case studies on integrated farming systems and development of IFS for different resource situations.

Successful Integrated farming by Sh. Rajbir Singh



Profile:

Name	:	Sh. Rajbir Singh
Father's Name	:	Sh. Umrao Singh
Village	:	Basirpur
Age	:	40 yr.
Land Area	:	28 Acre
Family members	:	9
Agri. Experience	:	10 yr.

Integrated farming is quite essential approach for maintaining sustainability in agriculture. This approach not only conserves soil & water resources but improve economic status of farmer by avoiding risk. Majority of farmers in the district inclined towards single component system which resulted in depletion of the resources. Sh. Rajbir Singh s/o Sh. Umrao Singh of village Basipur, Distt. Mahendergarh is also one of these farmers. Despite being a big farmer (28 Acres), he could not meet the household needs as the income generated from crop production system was marginal. He used to cultivate pearlmillet & clusterbean in kharif and wheat, mustard and gram in rabi season. By growing these crops he could generate net returns of Rs. 8,56,500 /-. The crop wise net return obtained by cultivation of different crops in different season is given in table below:-

Income generated from traditional farming system.

Crop	Area (Acre)	Net returns(Rs.)
Bajra	12	92000
Guar	14	307500
Wheat	8	194000
Mustard	16	228000
Gram	2	35000
Total	-	856500

In order to generate more income with resources available, he approached to KVK Scientists. Scientists of KVK on analysing his situation such as available resources (land, water and labour) and marketing avenues suggested him to adopt integrated farming system. The Scientists provided him a model of integrated farming system comprising of crop production, orchard cultivation, dairy farming & fish cultivation which can be successfully implemented under his conditions. Instead of adopting all the enterprises, simultaneously he began with Ber orchard (2.5 Acres) and dairy farming with rearing of two buffalo and five cross-bred cows. After successful adoption of these enterprises he adopted rest of enterprises (fish cultivation, poultry farming) of the suggested model. At present to generate more income from the enterprises adopted he introduced intercrops in Ber orchard and value addition to dairy unit by introducing preparation of rasgulla. At present in the dairy unit he has having two murrah buffaloes, and 16 cross-bred cows. By adopting this integrated farming system model he has been able to generate quite good net returns and conservation of resources as the enterprises included in the system by him are supportive to each other. Crop production component meets the feed and fodder demand of family and dairy animals. Manure obtained from animal & poultry

farming is useful in improving the soil conditions and the fish cultivation water is utilized in applying irrigation at critical stages of crops. Drip irrigation system in orchard is useful in conserving water resources. Thus, Sh. Rajbir Singh has been able to conserve his valuable resources and generate very good net return obtained by adoption of IFS approach. The net returns obtained from various enterprises are being given in table below:-

Income generated from Integrated Farming System.

Enterprise	Area/No. of Animals/Unit	Net Income(Lacs)
Crop Production	25	8.5
Ber orchard+ Intercrop	2.5	3.0
Dairy Farming	18(2M+16HF)	5.0
Fish Cultivation	187x167x20ft. Pond size	0.80
Poultry Farming	5000 birds	1.5
Rasgullaplant	One unit	4.8
	-	Total= 18.8

His hunger of generating more income could not stop here. He again approached to KVK Scientists to suggest some new approaches to make crop production system more remunerative. Keeping in view the unexpected rising prices of cluster bean produce (Rs. 10,000/qt.), KVK Scientists advised him to grow this crop in more area. He raised the area under this crop from 14 acre to 25 acre and adopted improved production practices. He was able to produce 200 Qtl. of crop grain yield which provided him an additional income of Rs. 13, 75,000. At present the status of net return obtained by Sh. Rajbir Singh is given below:

Crop	**Area (Acre)**	**Net returns(Rs.)**

Guar	25	17,75,000
Wheat	8	194000
Mustard	16	228000
Gram	2	35000
Total	-	22,32,000

Enterprise	**Area/No. of Animals/Unit**	**Net Income(Lacs)**
Crop Production	25	22.32
Ber orchard+ Intercrop	2.5	3.0
Dairy Farming	18(2M+16HF)	5.0
Fish Cultivation	187x167x20ft. Pond size	0.80
Poultry Farming	5000 birds	1.5
Rasgullaplant	One unit	4.8
Total	-	37.42

(Source: KVK, Mahendergarh)

Model 1: IFS model for different resource situations of Haryana

Details of various components of IFS along with allotted area

Treatments	**Crop sequence**	**Area allotted (m^2)**
T_1	Cotton – wheat	2000
T_2	Mungbean – wheat (tall)	500
T_3	Pearl millet – mustard	2000
T_4	Sorghum (F) – wheat	2000
T_5	Sorghum (F) – berseem (F)	750
T_6	Sorghum (F) – Oat (F)	750
T_7	Clusterbean-potato-okra	500
T_8	Okra – palak – bottle gourd	500
T_9	Horticultural crop (lemon + guava) + field crop as intercrop	600

T_{10}	Vermicompost	50
T_{11}	Mushroom unit	50
T_{12}	Dairy unit	300
T_{13}	Boundary plantation with bael	-

The IFS model is being practiced over an area of one ha. The yield and returns varied markedly among different cropping systems. Vegetable component gave net return of Rs 10470/- from an area of 1000 m^2 and cotton-wheat gave net return of Rs. 22608/- from an area of 2000 m^2. Residues of vegetable crops were incorporated in situ whereas cotton sticks were removed from the field. Guava and lemon trees are yet to bear fruits. The income from animal component added from February 2014 is Rs. 14453. The net return from above IFS model was Rs.79672 ha^{-1} $year^{-1}$.

Model 2: IFS model for marginal farmers, under irrigated dry situation of Andhra Pradesh

Components of model: Crop-Livestock-Horticulture-Back yard poultry - vermicompost

Modules/ Enterprises identified	**Area (ha)**	**Treatments**
Crops	0.70	Cropping Systems:
		Rice-Maize
		Rice-Sunflower
		Maize-Castor
		Pigeonpea +Greengram-Groundnut
		Maize + Pigeonpea - Cowpea (Fodder)
		Hybrid Napier and Lucerne
Horticulture	0.20	
Fruits		Guava

Vegetables -4		Tomato, Green Chillies, Carrot, Cluster bean
Livestock		
Buffaloes -2	0.04	Graded Murrah breed
Goat-5 +1	0.015	Osmanabadi breed
Back yard Poultry		
Fowls (Hens & Cocks)-18 +2		Vanaraja/Rajasri breed
Vermi-compost unit	0.015	
Supporting activities[#]	0.03	
TOTAL	1.00	

Economics of IFS unit

Component	**Area (m^2)**	**Returns (Rs $plot^{-1}Yr^{-1}$)**		
		Cost	**Gross**	**Net**
Crop	6000	31414	57092 (20.92)	25678 (55.76)
Horticulture	2000	9848	8425 (3.09)	-1423 (-3.09)
Livestock	1350	185544	206319 (75.62)	20775 (45.12)
Others (composting- FYM)	650	--	1017 (0.37)	1017 (2.21)
Total	10000	226805	272852	46047

Models of other resource situations will also be studied in the class.

Exercise No 4. Calculation of economics of different farming systems models

Modules/ Enterprises	**Number**
Field crops	Cereals, pulses, oilseeds & Fodder)
Horticulture	Olericulture - 600 m^2 Floriculture - 300 m^2 Pomology - 500 m^2
▪ Cross breed Cows	2 Cross breed cows +2 calves
▪ Goat unit	10 Black Bengal + 1 male Jamunapari
▪ Poultry	White Leghorn layers – 50 Nos.
▪ Fishery (Polyculture)	1000 Nos.

Modules/ Enterprises identified	Number
Apiary	8 boxes of Italian spp.
Mushroom production unit	Capacity of 2 kg/ day
Biogas unit	Capacity of 2 m^3 / day
Vermi-compost unit	6 Nos.

Food requirement of a family of 5 members

Sl. No.	**Particulars**	**Requirement (g/ Adult/ day)**	**Requirement (kg/ Adult/ year)**	**Requirement of 5 family members (kg/ year)**
1	Cereals	600 gm	220	1100
2	Pulses	50 gm	19	95
3	Oilseeds	70 gm	25	125
4	Vegetables	350 gm	128	640
5	Fruits	125 gm	45	225
6	Milk	300 ml	110	550

7	Honey	5 ml	1.8	9.0
8	Meat (If milk is not taken)	65 gm	24	120
9	Tuber	100 gm	36	18
10	Eggs	-	180	900

Feed and fodder requirement of different components in farming system

Sl. No	**Component**	**Requirement (kg/ Adult/ day)**			**Requirement (quintals / year)**		
		Dry fodder	**Green fodder**	**Concen -trate**	**Dry fodder**	**Green fodder**	**Concen -trate**
1	Dairy (2 cows)	5.0	12.0	3.30	36.5	88.0	22.0
2	Calves (2)	1.0	3.0	0.50	7.3	22.0	4.0
3	Poultry (50 birds)	-	-	0.12	-	-	22.0
4	Goatry (10+1)	0.3	2.0	0.30	12.0	81.0	12.0
5	Kids (25)	0.15	1.0	0.10	13.7	92.0	9.0
6	Mushroom				18.5	-	1.8
Total		6.45	18.0	4.67	89.5	283.0	71.0

Average income from crop production

Farm produce	**Gross area m^2**	**% of total**	**Expected production / year (kg)**	**Gross returns (Rs./year)**
Crop production (Net area = 6900 m^2)	-	69.0	-	-
Cereals (Paddy, maize, wheat)	11,800	49.8	6150	61,500
Pulses (Green gram)	2900	12.2	333	6,660
Oilseeds (mustard)	1000	4.2	180	3,600
Tuber (potato)	2000	8.3	4000	20,000
Green fodders	6000	25.0	28700 @	21,525

(Sorghum, berseem, oats, hybrid napier, cowpea			Rs. 75/q	
Dry fodders (Wheat & bajra straw, maize stover etc.)	-	-	9,700 @ Rs. 200/q	19,400
Total A	23,700	-	-	1, 32,685

Average production and expected income from horticultural unit.

Fruits/ vegetables	**Number /area**	**Productivity (kg/unit)**	**Prodn. (kg)**	**Market whole sale price/kg**	**Total produce Value (Rs.)**
A Fruits	500				
i) Papaya		30kg /plant	1800	Rs.10/ kg	18,000
ii) Banana		20 kg /plant	1200	Rs.5/ kg	6,000
iii)Straw berry		3.0 kg / plant	720	Rs.60/ kg	43,200
iv) Cape gooseberry		2.0 kg / plant	360	Rs.30/ kg	10,800
Total A	-		4080		78,000
B. Vegetables					
Brinjal, Tomato, Cabbage, Okra, Veg.pea, etc	600	Av.150 q / ha in each season	2700	Rs.10/ kg	27,000
Total B	-	-	2700		27,000
C Floriculture					
Marigold, Vinca etc.	300	Av.50 q / ha in each season	450	Rs.10/ kg	4,500
Total C			450		4,500
Gross returns		-			1,09,500

Dairy Unit

Cross breed cows - 02 Calves + 2

1	Dairy shed	40m^2	Gross income (Rs./ year)
2	Milk production @10 litres /cow/day (lactation - 265 days)	5,300 litres/ year@ Rs.15/ litre	79,500
3	Sale of young calves	02 Nos. @ Rs. 3,000/ calf	6,000
4	Fresh cow dung @ 70kg/day	25,500 Kg/ year	-
5	Cow urine @ 40 litres/ day	14,600 lit./ year	-
6	Vermi compost	10,200 kg	-
Gross income from dairy			85,500

GOAT: an essential component of IFS for landless and marginal farmers

ATM card for farmers

Female	10 Black Bengal
Male	01 Jamunapari

Particulars	**Details**	**Gross income (Rs./ year)**
Goat shed	50m^2	
Purpose	Meat	
Av. no. of kids	50 kids/ year	
Mortality @ 10%	5 kids	
Total live weight kg / year	506.0 kg/year	
Dressed weight @ 70 %	354 @ 200/ kg	70,800
Total excreta (Vermi-compost)	5740 (2300) kg	-
Gross income from goatry		70,800

FISHERY: 1000 polyculture fingerlings to be added in 0.1 ha ponded water

Type of fish	Proportions	Expected annual production (kg)	Gross returns (Rs./year)	Remarks
Catla	300 (30%)	225 @ Rs. 60/ kg	13,500	Surface feeder Fast growing Average growth >1kg/yr
Rohu	300 (30%)	175	10,500	Column feeder, Grows well in deeper ponds, Average growth > 1Kg /yr.
Mirgal	200 (20%)	125	7,500	Bottom feeder Average growth is <1kg/yr
Grass carp	200 (20%)	175	10,500	Column feeder ,feeds on grasses available in the pond Average growth >1kg/yr
Total	1000	700	42,000	-
Pond silt to be recycled in crop field				11.0 t/ year

Poultry Unit:

50 White Leghorn layer birds to be sheltered over fish pond at one corner to meet the feed requirement of 1000 polyculture fingerlings in 0.1 ha ponded water

Sl. No.	Particulars	Details	Gross returns (Rs./year)
1	Poultry shed	$25m^2$	-
2	Number of birds	50	-
3	Egg production @ 250/bird/year	12,500 eggs /year @ Rs2.50/	31,250

		egg	
4	Meat after 72 weeks @ 1.7 kg /bird	85 kg @ Rs. 60/ kg	5,100
5	Poultry droppings in fish pond	1640 kg / year	-
Gross income from poultry unit		-	36,350

Apiary (Bee keeping)

- A small unit with 8 honey bee boxes of Italian species (*Apis mellifera*) will be kept
- Bees will be helpful in increasing yield and quality of produce through improvement in pollination.
- It is a very good supplementary enterprise in farming system, because it does not require any raw material. Nectar and pollen flowers are raw materials which is available in plenty in nature.
- Total honey production- 320 litres /year @ 40 kg / box/ year
- Gross returns (Rs./year) - 320 litres /year @ Rs. 160/ litre = Rs.51,200

Biogas plant

Capacity- 2 m^3 biogas/ day

- Biogas is a clean, unpolluted and cheap source of energy
- Biogas to be used for cooking and lighting.
- Biogas is also to be used for lighting the fish pond to attract insects which form a choice food for fishes.
- 60 kg cow dung or 44 kg goat dropping / day will be required to produce 2 m^3 gas/ day
- Slurry = 57 or 41 kg/ day will go to Vermicompost
- Total production - 730 m^3 (2 m^3 = 1.5 lit.

Kerosene-548 lit. @ Rs.12/ litre

- Gross returns (Rs./year) - 6,570

Mushroom production

Types	**Oyster**
Production	2 kg /day
No. of beg/beds to be required	1850
Paddy/ wheat straw	1850 kg /Year
Pulse powder	185 kg
Mushroom production	730 kg @ Rs.50/ kg
Gross income from mushroom	Rs. 36,500

Exercise No 5. Visit to different units of IFS in different agro climatic zones of Haryana

Visit to different units of IFS in different agro climatic zones of Haryana will be conducted and the students will interact with the unit holder / farmer and will gather the following information on different components of IFS units

Crop: Students will collect information on various aspects of crop production from the farmer will record and interpret in their own words:

Poultry:

1. Year of start of poultry unit=
2. Information regarding house hold members=
3. Number of birds : Male = Female =
4. Weight of mature bird =
5. Total egg production / year =
6. Total meat production / year =
7. Total income =
8. Inputs supplied to bird =
9. Inputs supplied from the system =
10. Inputs purchased from outside farm =
11. Cost of inputs purchased from outside =
12. Household consumption of eggs =
13. Quantity of manures provided to other enterprises on the farm =
13. Household consumption of meat =
14. Household consumption of eggs =
15. Net returns from the poultry unit =

Dairy:

1. No. of animals =
2. Breed of animals =
3. Average age of animals =
4. Purchase value of animals =

Input flow for dairy farming

Sr. No.	Particulars	Quantity utilized (g)		Price (Rs.)
		On farm produce	Purchased from market	
a.	Grain			
b.	Green fodder			
c.	Crops byproducts			

External inputs utilized:

Sr.No.	Particulars	Quantity/ Unit	Price (Rs.)/Unit	Total Cost (Rs.)
a.	Concentrate feed			
b.	Agro industrial by products			
c.	Medicines			
d.	Mineral mixture			
e.	Vaccine			
f.	Labour (man days/year)			

Output flow:

Sr.No.	Particulars	Quantity/ Unit	Price (Rs.)/Unit	Total Cost (Rs.)
a.	Average daily milk production			
b.	Annual milk production			
c.	Milk used for home consumption			
d.	Milk sold in market			

1. Quantity of dung produced (annual) =
2. Quantity of dung used in vermicompost =

3. Quantity of dung used in FYM =
4. Income from sale of vermicompost (Rs.) =
5. Income from sale of FYM (Rs.) =
6. Quantity of dung used for fuel =
7. Quantity of vermicompost used in the self farm =
8. Quantity of FYM used for self farm =

Calf production :

1. Total no. of calves =
2. Age of calf at the time of sale =
3. Sale price of calf =

Income :

1. Gross return (Rs.) =
2. Net return (Rs.) =
3. B : C =